

# On Ramanujan Challenge Problem 3.2: The Apéry GCD Conjecture

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## Abstract

We study  $G_n = \gcd(d_n a_n, d_n b_n)$  for the Apéry sequences  $(a_n), (b_n)$  with  $d_n = \text{lcm}(1, \dots, n)^3$ . We prove *unconditionally* that  $G_n = e^{o(n)}$  for a set of positive integers of natural density 1, with the quantitative bound  $\#\{n \leq N : \log G_n > \varepsilon n\} = O_\varepsilon((\log N)^2)$  for every  $\varepsilon > 0$ ; in particular, the exceptional set has upper Banach density zero and finite harmonic weight. The key ingredient is a new bound  $Z(p) = O(p^{2/3})$  on the zero-count function  $Z(p) = \#\{j < p : p \mid b_j\}$ , derived from the Apéry recurrence via a gap polynomial argument. Computation for all 78,496 primes  $5 \leq p \leq 10^6$  gives  $Z(p) \leq 12$  with mean  $\approx 1$ . The pair count  $Z(p)/2$  fits Poisson(1/2); we prove this rigorously in the prime-aspect via a Chebotarev argument for the hyperoctahedral Galois group of the gap polynomials. The Poisson tail predicts  $\max_{p \leq N} Z(p) \sim 2 \log \log N$ , so  $Z(p)$  is almost certainly unbounded. Under the weaker Cesàro hypothesis  $\sum_{p \leq x} Z(p) \ll \pi(x)$  (Hypothesis  $\bar{Z}$ ), which the Poisson model supports, we prove  $\log G_n = O(\sqrt{n})$  for density 1, matching the unconditional small-prime floor, with the conditional exceptional set shrinking to  $O(N^{1/3})$  and having upper Banach density zero. The proof combines a dyadic divisor-incidence bound with a sub-interval zero-count argument to reduce the leading-digit bottleneck from  $O(N^{3/2})$  to  $O(N^{10/7}/\log N)$  unconditionally, and to  $O(N^{4/3}/\log N)$  under Hypothesis  $\bar{Z}$ .

## 1 Setup

The Apéry sequences satisfy the three-term recurrence

$$(n+1)^3 u_{n+1} = (34n^3 + 51n^2 + 27n + 5) u_n - n^3 u_{n-1}, \quad n \geq 1, \quad (1)$$

with  $(a_0, a_1) = (0, 6)$  and  $(b_0, b_1) = (1, 5)$ . Here  $b_n = \sum_{k=0}^n \binom{n}{k}^2 \binom{n+k}{k}^2 \in \mathbb{Z}$  is the  $n$ -th Apéry number. Set  $d_n = \text{lcm}(1, \dots, n)^3$ ; Apéry [3] showed  $d_n a_n, d_n b_n \in \mathbb{Z}$ . Define  $G_n = \gcd(d_n a_n, d_n b_n)$ . Write  $P(n) = 34n^3 + 51n^2 + 27n + 5$  for the middle coefficient.

## 2 The Wronskian identity

**Lemma 1.** *For all  $n \geq 1$ ,  $W_n := a_n b_{n-1} - a_{n-1} b_n = 6/n^3$ .*

*Proof.* The Casorati determinant satisfies  $W_{n+1}/W_n = n^3/(n+1)^3$  from (1). Since  $W_1 = a_1 b_0 - a_0 b_1 = 6$ , telescoping gives  $W_n = 6/n^3$ .  $\square$

**Corollary 2.** *For every prime  $p \geq 5$ ,*

$$v_p(G_n) \leq 3 \lfloor \log_p n \rfloor. \quad (2)$$

*Proof.*  $G_n$  divides  $d_n(a_n b_{n-1} - a_{n-1} b_n) = 6d_n/n^3$ . For  $p \geq 5$ ,  $v_p(6) = 0$  and  $v_p(d_n/n^3) = 3 \lfloor \log_p n \rfloor - 3v_p(n) \leq 3 \lfloor \log_p n \rfloor$ .  $\square$

### 3 Small primes: unconditional bound

**Proposition 3.**  $\sum_{p \leq \sqrt{n}} v_p(G_n) \log p = O(\sqrt{n}).$

*Proof.* By (2),  $v_p(G_n) \log p \leq 3 \log n$  for every prime  $p \geq 5$ . Primes  $p = 2, 3$  contribute at most  $O(\log n)$  each. There are  $O(\sqrt{n}/\log n)$  primes up to  $\sqrt{n}$ , so the sum is at most  $3 \log n \cdot O(\sqrt{n}/\log n) = O(\sqrt{n})$ .  $\square$

### 4 The multiplicative Lucas congruence

**Theorem 4** (Gessel [13]). *For every prime  $p$  and  $n = \sum_{i=0}^s n_i p^i$  in base  $p$ ,*

$$b_n \equiv \prod_{i=0}^s b_{n_i} \pmod{p}. \quad (3)$$

This follows from the combinatorial identity  $b_n = \sum_k \binom{n}{k}^2 \binom{n+k}{k}^2$  and termwise application of Lucas's congruence to each binomial coefficient. See also Mimura [17] and McIntosh [16].

### 5 The zero-count function

**Definition 5.** For a prime  $p \geq 5$ , define  $Z(p) = \#\{j \in \{0, \dots, p-1\} : b_j \equiv 0 \pmod{p}\}$ .

#### 5.1 Unconditional bound

**Lemma 6** (No consecutive zeros). *For every prime  $p \geq 5$  and every  $j \in \{0, \dots, p-2\}$ ,  $b_j$  and  $b_{j+1}$  cannot both vanish modulo  $p$ .*

*Proof.* Suppose  $b_m \equiv b_{m+1} \equiv 0 \pmod{p}$  with  $0 \leq m \leq p-2$ . The recurrence (1) at  $n = m$  gives  $(m+1)^3 b_{m+1} = P(m) b_m - m^3 b_{m-1}$ , so  $m^3 b_{m-1} \equiv 0 \pmod{p}$ . Since  $1 \leq m \leq p-2$  implies  $p \nmid m$ , we get  $b_{m-1} \equiv 0$ . Iterating gives  $b_0 \equiv 0 \pmod{p}$ , contradicting  $b_0 = 1$ .  $\square$

**Lemma 7** (Gap polynomial). *Fix  $h \geq 2$ . Suppose  $b_m \equiv 0 \pmod{p}$  and  $m+h \leq p-1$ . Then*

$$b_{m+h} \equiv C_h(m) b_{m+1} \pmod{p}, \quad (4)$$

where  $C_h(m)$  is a rational function of  $m$  whose numerator (after clearing the denominator  $\prod_{j=1}^{h-1} (m+j+1)^3$ ) is a polynomial of degree exactly  $3(h-1)$  with positive leading coefficient.

In particular, if  $b_m \equiv b_{m+h} \equiv 0 \pmod{p}$ , then by Lemma 6  $b_{m+1} \not\equiv 0$ , so  $C_h(m) \equiv 0 \pmod{p}$ , and  $m$  is constrained to at most  $3(h-1)$  residue classes modulo  $p$ , provided  $C_h$  is not identically zero over  $\mathbb{F}_p$ .

*Proof.* We iterate the recurrence from the initial condition  $(b_m, b_{m+1}) = (0, b_{m+1})$ . At step  $n = m+1$ :  $b_{m+2} = P(m+1) b_{m+1} / (m+2)^3$ , since  $b_m = 0$  kills the second term. At step  $n = m+j$  for  $j \geq 2$ :  $b_{m+j+1}$  is a  $\mathbb{Q}(m)$ -linear combination of  $b_{m+j}$  and  $b_{m+j-1}$ , both already rational multiples of  $b_{m+1}$ . The leading coefficient of  $P(n)$  is 34 and the subtracted term has leading coefficient 1, so at each step the degree in  $m$  increases by 3 and the leading coefficient remains positive. After  $h-1$  steps the numerator has degree  $3(h-1)$ .

For  $h = 2$ :  $C_2(m) = P(m+1)/(m+2)^3$ , with numerator  $P(m+1) = (2m+3)(17m^2+51m+39)$ , a cubic with leading coefficient 34.  $\square$

**Lemma 8** (Nonvanishing over  $\mathbb{F}_p$ ). *For every prime  $p \geq 7$  and every  $2 \leq h \leq p$ , the numerator of  $C_h$  is not the zero polynomial modulo  $p$ .*

*Proof.* Write  $N_h$  for the numerator of  $C_h$  (a polynomial of degree  $3(h-1)$ ), satisfying  $N_{h+1}(m) = P(m+h)N_h(m) - (m+h)^6 N_{h-1}(m)$ , with  $N_1 = 1$  and  $N_2 = P(m+1)$ .

*First evaluation.* Equation (4) at  $m = -1$  (where  $b_{-1} = 0$ ,  $b_0 = 1$ ) gives  $C_h(-1) = b_{h-1}$ , hence  $N_h(-1) = b_{h-1} \cdot ((h-1)!)^3$ .

*Second evaluation.* Setting  $m = -2$  in the recurrence gives  $N_{h+1}(-2) = P(h-2)N_h(-2) - (h-2)^6 N_{h-1}(-2)$ , with  $N_1(-2) = 1$  and  $N_2(-2) = P(-1) = -5$ . By induction using the Apéry recurrence at  $n = h-2$ :

$$N_h(-2) = -5b_{h-2}((h-2)!)^3 \quad (h \geq 2). \quad (5)$$

*Conclusion.* If  $N_h \equiv 0 \pmod{p}$  as a polynomial ( $2 \leq h \leq p$ ,  $p \geq 7$ ), then  $N_h(-1) \equiv N_h(-2) \equiv 0$ , forcing  $b_{h-1} \equiv b_{h-2} \equiv 0 \pmod{p}$  (since  $p \nmid 5$  and  $p \nmid (h-1)!, (h-2)!$ ). These are consecutive zeros, contradicting Lemma 6.  $\square$

**Proposition 9.** *For every prime  $p \geq 7$ ,*

$$Z(p) \leq \left\lfloor \frac{p}{H} \right\rfloor + \frac{3}{2}(H-1)(H-2) + 1, \quad (6)$$

where  $H$  is any integer with  $2 \leq H \leq p$ . Optimizing  $H = \lfloor (p/3)^{1/3} \rfloor + 1$  gives  $Z(p) \leq \frac{3^{4/3}}{2} p^{2/3} + O(p^{1/3}) = O(p^{2/3})$  unconditionally for all primes. In particular,  $Z(p) = o(p)$ .

*Proof.* Partition  $\{0, \dots, p-1\}$  into  $\lceil p/H \rceil$  consecutive blocks of length at most  $H$ . In each block, mark the first zero; all other zeros have gap  $h \in \{2, \dots, H-1\}$  to some earlier zero in the same block (gap 1 is impossible by Lemma 6). By Lemma 7, a gap- $h$  pair constrains  $m$  to at most  $3(h-1)$  residue classes mod  $p$  (the denominator factors  $(m+j+1)^3$  are nonzero since  $0 \leq m+j+1 \leq p-1$ ). By Lemma 8,  $C_h \not\equiv 0$  over  $\mathbb{F}_p$ , so the numerator roots are at most  $3(h-1)$ . Summing:

$$Z(p) \leq \left\lfloor \frac{p}{H} \right\rfloor + \sum_{h=2}^{H-1} 3(h-1) = \frac{p}{H} + \frac{3}{2}(H-1)(H-2) + O(1).$$

The minimum over real  $H > 0$  occurs at  $H = (p/3)^{1/3}$ , giving the bound. For  $p = 2, 3, 5$ :  $Z(p) = 0$  (since  $b_j$  for  $j < 5$  is coprime to  $2, 3, 5$ ).  $\square$

**Corollary 10** (Sub-interval zero count). *For every prime  $p \geq 7$  and every interval  $I \subseteq \{0, \dots, p-1\}$ ,*

$$\#\{j \in I : p \mid b_j\} = O(|I|^{2/3}).$$

*Proof.* Apply the proof of Proposition 9 to the sub-interval  $I$  in place of  $\{0, \dots, p-1\}$ : partition  $I$  into  $\lceil |I|/H \rceil$  blocks of length  $H$ , bound the non-first zeros in each block by the same gap polynomial argument (Lemma 7 applies regardless of the interval), and optimize.  $\square$

*Remark 11.* The leading coefficient of the numerator of  $C_h$  is  $U_{h-1}(17)$ , the Chebyshev polynomial of the second kind evaluated at 17, satisfying  $\ell_1 = 1$ ,  $\ell_{h+1} = 34\ell_h - \ell_{h-1}$ . Let  $N_h(m)$  denote the cleared numerator of  $C_h$ . The identity  $P(-u-1) = -P(u)$  (evident from  $P(n) = (2n+1)(17n^2 + 17n + 5)$ ) propagates through the recurrence to give  $N_h(-m-h-1) = (-1)^{h-1} N_h(m)$ , verified computationally for  $h \leq 200$ . For even  $h$ ,  $N_h$  is an *odd* polynomial in  $(m + (h+1)/2)$ , forcing the structural factor  $(2m+h+1)$ . At  $h = 2$ : factor  $(2m+3)$  in  $P(m+1) = (2m+3)(17m^2 + 51m + 39)$ . The computational data (Section 10) suggests  $Z(p) = O(1)$ , far stronger than  $O(p^{2/3})$ .

*Remark 12* (Content restriction). A stronger nonvanishing holds: if  $p \geq 7$  divides the content of  $N_h$  (the gcd of all its coefficients), then  $h \geq 2p$ . Indeed,  $N_h(-r) = (-1)^{r-1} b_{r-1} b_{h-r} ((r-1)!)^3 ((h-r)!)^3$  for  $1 \leq r \leq h$  (by induction from the Apéry recurrence); choosing  $r = 1, 2$  and using  $p \nmid (h-1)!, (h-2)!, 5$  for  $h \leq p$  recovers Lemma 8, while the range  $p < h < 2p$  is excluded similarly via  $r = h - p + 1$ . Setting  $\beta_n = (n!)^3 b_n$  and  $F(z) = \sum_{n \geq 0} b_n z^n$ , the endpoint factorization packages into the double generating function  $\sum_{r,s \geq 0} \frac{N_{r+s+1}(-r-1)}{(r!)^3 (s!)^3} u^r v^s = F(-u) F(v)$ , showing that the boundary array is a rank-one tensor.

*Remark 13* (Adjacent resultant formula). The adjacent resultants satisfy the exact formula

$$|\text{Res}(N_h, N_{h+1})| = \prod_{j=1}^{h-1} (j!^3 b_j)^6. \quad (7)$$

This follows from the recursion  $\text{Res}(N_h, N_{h+1}) = N_h(-h)^6 \text{Res}(N_{h-1}, N_h)$  (since  $N_{h+1} \equiv -(m+h)^6 N_{h-1} \pmod{N_h}$ ), combined with  $N_h(-h) = (-1)^{h-1} N_h(-1) = (-1)^{h-1} ((h-1)!)^3 b_{h-1}$  (the reflection identity and the first evaluation from Lemma 8). Verified for  $h = 2, 3, 4, 5$ ; the cases  $h = 2, 3, 4$  give  $\text{Res}(N_2, N_3) = -5^6$ ,  $\text{Res}(N_3, N_4) = -2^{18} \cdot 5^6 \cdot 73^6$ ,  $\text{Res}(N_4, N_5) = 2^{36} \cdot 3^{18} \cdot 5^{12} \cdot 17^{12} \cdot 73^6$ .

A key consequence:  $p \mid \text{Res}(N_h, N_{h+1})$  if and only if  $p \mid b_j$  for some  $1 \leq j \leq h-1$  (when  $p > h$ , so factorials are coprime to  $p$ ). In particular,  $N_2$  and  $N_3$  have disjoint root sets modulo  $p$  for all  $p \geq 7$  (since  $5^6 \neq 0$ ).

*Remark 14* (Sharpness of the bound). The exponent  $2/3$  in Proposition 9 is optimal for any argument using only the degree bounds  $\deg N_h = 3(h-1)$ . The exact extremal problem  $\max\{\sum r_h : r_h \leq 3(h-1), \sum h r_h \leq p\}$  has value  $\frac{3}{2} p^{2/3} + O(p^{1/3})$ , with leading constant  $3/2$  achieved by filling gap sizes  $h = 2, 3, \dots, H$  to capacity and using the remaining space at  $h = H+1$ , where  $H \sim p^{1/3}$ . Moreover, degree bounds alone (even using all pairs of zeros, not just consecutive pairs) cannot yield  $O(p^{1/2})$ : an abstract polynomial model on  $\sim p^{2/3}$  points can satisfy  $\deg F_h \leq 3(h-1)$  for all relevant  $h$ . Improving the exponent requires genuinely arithmetic input about the family  $(N_h)$ .

*Remark 15* (Dodgson condensation). The Desnanot–Jacobi identity (Lewis Carroll’s condensation) applied to the tridiagonal determinant gives the bilinear relation

$$N_{h+2}(m) N_h(m+1) - N_{h+1}(m) N_{h+1}(m+1) = - \prod_{j=2}^{h+1} (m+j)^6. \quad (8)$$

This is an identity in  $\mathbb{Z}[m]$ , verified for  $h \leq 7$ . A consequence: for  $m \not\equiv -j \pmod{p}$  ( $2 \leq j \leq h+1$ ), the pairs  $(m, h+2)$  and  $(m+1, h)$  cannot both lie in the zero relation  $\mathcal{Z} = \{(m, h) : N_h(m) \equiv 0\}$ , since  $N_{h+1}(m) N_{h+1}(m+1)$  would equal a nonzero product of sixth powers. After a meromorphic gauge  $\tau_h(x) = N_h(x)/d_h(x)$  satisfying  $d_{h+2}(x) d_h(x+1) = \prod_{j=2}^{h+1} (x+j)^6$ , this becomes an exact  $SL_2$ -tiling:  $\tau_{h+1}(x) \tau_{h+1}(x+1) - \tau_{h+2}(x) \tau_h(x+1) = 1$ , equivalently an  $A_1$   $T$ -system (discrete Liouville equation). A zero of  $\tau$  forces its transverse neighbors to be units (*singularity confinement*), but abstract  $SL_2$ -tilings can have zeros of density  $1/2$  (e.g.,  $q_n = 0, 1, 0, -1, \dots$  with  $q_n^2 - q_{n-1} q_{n+1} = 1$ ), so the bilinear structure alone does not force a sublinear zero count. The arithmetic content—endpoint factorization, restart identity (Lemma 21), and polynomial degree bounds—is essential.

*Remark 16* (Projective orbit reformulation). Let  $B_n = n!^3 b_n$  and  $D_n$  be the second solution of the renormalized recurrence  $w_{n+1} = P(n) w_n - n^6 w_{n-1}$  with  $(D_0, D_1) = (0, 1)$ . Their Casoratian is  $B_n D_{n+1} - D_n B_{n+1} = (n!)^6$ , nonzero for  $n < p$ . Define the *projective Apéry orbit*  $\pi : \{0, \dots, p-1\} \rightarrow \mathbb{P}^1(\mathbb{F}_p)$  by  $\pi_n = [B_n : D_n]$ . For  $0 \leq m < m+h < p$ ,  $N_h(m) \equiv 0 \pmod{p}$  if and only if  $\pi_m = \pi_{m+h}$ . Thus the zero-count  $Z(p)$  equals the number of distinct collisions in the orbit  $\pi$ . The reflection

symmetry  $P(-1-n) = -P(n)$  implies  $b_{p-1-n} \equiv b_n \pmod{p}$ , making the orbit a palindrome:  $\pi_n = \pi_{p-1-n}$ . In particular,  $Z(p)$  is even except at the non-ordinary primes  $p \mid a_p(f)$  (at most two primes  $p \leq 10^5$ ).

*Remark 17* (Galois structure). For each fixed  $h$ , the Galois group of the splitting field of  $N_h$  over  $\mathbb{Q}$  is not the full symmetric group: the reflection  $N_h(-m-h-1) = (-1)^{h-1} N_h(m)$  forces roots into pairs  $\{r, -r-h-1\}$ , and  $\text{Gal}(N_h) \cong C_2 \wr S_m$  (hyperoctahedral group) with  $m = \lfloor 3(h-1)/2 \rfloor$  (confirmed exactly for  $h \leq 5$ ,  $2 \leq h \leq 9$  by obstruction tests). By the Chebotarev density theorem, the average number of  $\mathbb{F}_p$ -roots of  $N_h$  is 1 for odd  $h$  and 2 for even  $h$ . This prediction is verified: for 262 primes in  $[1000, 3000]$ , the measured averages are 0.98, 2.03, 1.11, 2.07, 1.11, 1.88 for  $h = 3, \dots, 8$ .

**Proposition 18** (Hyperoctahedral Poisson law). *Assume  $\text{Gal}(N_h) \cong C_2 \wr S_m$  with  $m = \lfloor 3(h-1)/2 \rfloor$ . Let  $Y_h(p)$  be the number of non-central reflection-pairs of roots of  $N_h$  that split completely over  $\mathbb{F}_p$ . Then for each fixed  $k \leq m$ , the factorial moments in the prime-aspect satisfy*

$$\lim_{x \rightarrow \infty} \frac{1}{\pi(x)} \sum_{p \leq x} (Y_h(p))_k = \left(\frac{1}{2}\right)^k,$$

where  $(Y)_k = Y(Y-1) \cdots (Y-k+1)$ . Since these match the factorial moments of  $\text{Poisson}(1/2)$  for all  $k$ , the prime-aspect distribution of  $Y_h(p)$  converges to  $\text{Poisson}(1/2)$  as  $h \rightarrow \infty$ .

*Proof.* In  $G = C_2 \wr S_m = \{(\sigma, \varepsilon) : \sigma \in S_m, \varepsilon \in \{\pm 1\}^m\}$ , a positive fixed point of  $g = (\sigma, \varepsilon)$  is an index  $i$  with  $\sigma(i) = i$  and  $\varepsilon_i = +1$ . The Frobenius at  $p$  acts on the  $m$  reflection-pairs, and  $Y_h(p)$  equals the number of positive fixed points. For any  $k$  distinct indices  $i_1, \dots, i_k$ , the fraction of  $g \in G$  fixing all  $k$  positively is

$$\frac{2^{m-k} (m-k)!}{2^m m!} = \frac{1}{2^k (m)_k}.$$

Summing over all  $(m)_k$  ordered  $k$ -tuples,  $\mathbf{E}_{g \in G}[(Y(g))_k] = (1/2)^k$ , which matches the factorial moments of  $\text{Poisson}(1/2)$ . The Chebotarev density theorem then gives the prime-aspect convergence (Remark 20 supplies the effective error term under GRH). Since the factorial moments determine the Poisson distribution and they hold for all  $k \leq m$ ,  $Y_h(p) \rightarrow \text{Poisson}(1/2)$  as  $m \rightarrow \infty$ .  $\square$

*Remark 19* (Prime-aspect vs. orbit-aspect). Proposition 18 is a *prime-aspect* statement: for fixed  $h$  and varying  $p$ , the root-pair count  $Y_h(p)$  converges to  $\text{Poisson}(1/2)$ . The zero-count  $Z(p)$  aggregates contributions from all gaps  $h = 1, \dots, p-1$  at a *fixed* prime  $p$ , so the orbit-aspect independence needed to deduce  $Z(p)/2 \sim \text{Poisson}(1/2)$  is not implied by Chebotarev alone. The gap between the two aspects parallels the gap between the Sato–Tate distribution (average over primes) and pointwise distribution (a single prime).

*Remark 20* (GRH-effective error). For each fixed  $k$ , the  $k$ -th factorial moment does not require the full splitting field  $K_h$  (of degree  $2^m m!$ , double-exponential in  $h$ ). Let  $E_{h,k}$  be the fixed field of the stabilizer of an ordered signed  $k$ -tuple with distinct underlying reflection pairs; then  $[E_{h,k} : \mathbb{Q}] = 2^k (m)_k = O_k(h^k)$ . Under GRH for  $\zeta_{E_{h,k}}$ ,

$$\frac{1}{\pi(x)} \sum_{\substack{p \leq x \\ p \text{ good}}} (Y_h(p))_k = 2^{-k} + O_k\left(\frac{\log D_{h,k} + 2^k (m)_k \log x}{\sqrt{x}}\right),$$

where  $D_{h,k} = |\text{Disc}(E_{h,k})|$ . The Apéry recurrence gives the coefficient-height bound  $\log \|N_h\|_1 = O(h \log h)$ , from which  $\log D_{h,k} \ll_k h^{k+2} \log h$ . Thus an  $O(h^{-C})$  error holds in the polynomial range  $x \geq h^{2k+4+2C+\varepsilon}$ , for each fixed  $k$ .

At  $x = 10^6$  with  $h = 10$  ( $m = 13$ ), the degree contributions alone for  $k = 1, 2, 3$  are 0.25, 5.99, and 131.7 respectively: the observed Poisson accuracy is far better than present effective Chebotarev can certify.

## 5.2 Dual bounds: the restart identity and column estimate

The bound  $Z(p) = O(p^{2/3})$  controls zeros along a *fixed row* (varying  $m$  for a given gap  $h$ ). A dual bound controls zeros along a *fixed column* (varying  $h$  for a given starting point  $x_0$ ).

**Lemma 21** (Restart identity). *Suppose  $N_r(x_0) = 0$  and  $N_{r+1}(x_0) \neq 0$  for some  $x_0 \in \mathbb{F}_p$  and  $r \geq 1$ . Then for every  $d \geq 0$ ,*

$$N_{r+d}(x_0) = N_{r+1}(x_0) \cdot N_d(x_0 + r). \quad (9)$$

*In particular, if  $N_{r+d}(x_0) = 0$  as well, then  $N_d(x_0 + r) = 0$ .*

*Proof.* The sequence  $Y_d = N_{r+d}(x_0)/N_{r+1}(x_0)$  has  $Y_0 = 0$ ,  $Y_1 = 1$ , and satisfies  $Y_{d+1} = P((x_0+r)+d) Y_d - ((x_0+r)+d)^6 Y_{d-1}$ , which is the defining recurrence for  $N_d(x_0 + r)$ .  $\square$

**Proposition 22** (Column bound). *For every  $x_0 \in \mathbb{F}_p$  and every interval  $I$  of  $H$  consecutive  $h$ -values with  $H \leq p$ ,*

$$\#\{h \in I : N_h(x_0) \equiv 0 \pmod{p}\} \leq \frac{3^{4/3}}{2} H^{2/3} + O(H^{1/3} + 1). \quad (10)$$

*Proof.* Split  $I$  at the unique transition  $h$  where  $x_0 + h \equiv 0 \pmod{p}$  (if it exists in  $I$ ), creating at most two nonsingular pieces plus an  $O(1)$  additive cost.

On a nonsingular piece, order the zero levels. For threshold  $L \geq 2$ : a gap of length  $d \leq L$  between consecutive zeros at levels  $r < r + d$  gives  $N_d(x_0 + r) = 0$  by Lemma 21. For fixed  $d$ , there are at most  $3(d-1)$  values of  $x_0 + r$  satisfying this (by Lemma 8 and  $\deg N_d = 3(d-1)$ ). Gaps longer than  $L$  contribute at most  $H/(L+1)$  zeros. Summing:

$$\text{zero count} \leq \frac{H}{L+1} + \sum_{d=2}^L 3(d-1) + O(1) = \frac{H}{L+1} + \frac{3}{2}L(L-1) + O(1).$$

Optimizing  $L \asymp (H/3)^{1/3}$  gives the stated bound.  $\square$

Computationally, the column count at mesoscopic scale  $H = \lfloor \sqrt{p} \rfloor$  is far below the  $O(H^{2/3})$  bound: for all primes  $p \leq 100,003$  and all  $x_0$ ,  $\#\{h \leq H : N_h(x_0) \equiv 0\} \leq 2$ . For  $p \geq 10,007$ , the maximum is  $\leq 1$ .

**Remark 23** (Rectangular incidence estimate). The row bound  $|R_h| \leq 3(h-1)$  and the column bound (10) together give the rectangular incidence estimate

$$\#\{(x, h) : x \in \mathbb{F}_p, 2 \leq h \leq H, N_h(x) \equiv 0\} \leq \min\{H^2, pH^{2/3}\}.$$

The first bound sums row degrees; the second sums column estimates. They cross at  $H \sim p^{3/4}$ . Note that a uniform bound  $R_p(H) \leq CH$  for all primes is *false*: by the Chebotarev density theorem, for each fixed  $H$ , infinitely many primes  $p > H^2$  have every  $N_h$  ( $2 \leq h \leq H$ ) splitting completely over  $\mathbb{F}_p$ , giving  $R_p(H) = \frac{3}{2}H(H-1)$ . The correct target is the *mesoscopic* estimate  $R_p(\lfloor \sqrt{p} \rfloor) = O(\sqrt{p})$ , where the split-prime obstruction does not apply because  $H = \sqrt{p}$  grows with  $p$ . More generally, if  $R_p(H) \ll H^\beta$  then  $Z(p) \leq p/H + R_p(H)$  is optimized at  $H \sim p^{1/(\beta+1)}$ , giving  $Z(p) \ll p^{\beta/(\beta+1)}$ : the current  $\beta = 2$  gives  $2/3$  and  $\beta = 1$  yields  $1/2$ . Reflection forces  $R_p(H) \geq H/2$  from central roots, so  $\beta = 1$  is the intrinsic limit of this first-moment method.

### 5.3 The continuant Bezout identity

Write  $S_d(x) = \prod_{j=2}^d (x+j)$ .

**Proposition 24** (Bezout identity). *For integers  $2 \leq d < e$ ,*

$$N_{e-1}(x+1) N_d(x) - N_{d-1}(x+1) N_e(x) = S_d(x)^6 N_{e-d}(x+d). \quad (11)$$

*In particular, after saturating the cut-edge divisor  $S_d$ ,*

$$(N_d, N_e) : S_d^\infty = (N_d, N_{e-d}(x+d)) : S_d^\infty.$$

*Proof.* The continuant addition formula (splitting the interval  $[1, e-1]$  at position  $d$ ) gives  $N_e = N_d N_{e-d+1}(x+d-1) - (x+d)^6 N_{d-1} N_{e-d}(x+d)$ . The Dodgson identity at index  $d-2$  gives  $N_{d-1}(x+1) N_{d-1} - N_{d-2}(x+1) N_d = \prod_{j=2}^{d-1} (x+j)^6$ . Combining these two identities yields (11). Verified computation-ally for all  $2 \leq d < e \leq 9$ .  $\square$

**Corollary 25** (Separated-block criterion). *Away from the cut-edge divisor  $S_d(x) = 0$ ,*

$$N_d(\alpha) \equiv N_e(\alpha) \equiv 0 \pmod{p} \iff N_d(\alpha) \equiv N_{e-d}(\alpha+d) \equiv 0 \pmod{p}.$$

*In particular,  $\deg \gcd_{\mathbb{F}_p}(N_d, N_e) \leq 3(\min(d, e-d) - 1)$ , which improves the naive  $3(d-1)$  when  $e-d < d$ .*

**Remark 26** (Separated-block resultant). Define  $\mathcal{R}_{d,r} = \text{Res}_x(N_d(x), N_r(x+d))$ . The Bezout identity gives

$$|\text{Res}(N_d, N_e)| = \prod_{j=1}^{d-1} ((j!)^3 b_j)^6 \cdot |\mathcal{R}_{d,e-d}|.$$

For adjacent indices ( $e = d+1$ ),  $N_1 = 1$  gives  $\mathcal{R}_{d,1} = 1$ , recovering the adjacent resultant formula. For nonadjacent indices,  $\mathcal{R}_{d,r}$  is genuinely new. The reflection identity gives  $|\mathcal{R}_{d,r}| = |\mathcal{R}_{r,d}|$ . The subtraction step does *not iterate*: after one application,  $N_d(x)$  and  $N_{e-d}(x+d)$  live on disjoint shifted intervals with opposite orientations, so no second nested division is possible. For  $d = 2$ , writing  $P(y) = (2y+1)Q(y)$  with  $Q(y) = 17y^2 + 17y + 5$  and  $\theta$  a root of  $Q$ , the norm decomposition  $\mathcal{R}_{2,r} = 2^{3(r-1)} G_r(-\frac{1}{2}) \cdot 17^{3(r-1)} N_{\mathbb{Q}(\theta)/\mathbb{Q}}(G_r(\theta))$  (where  $G_r = N_r(x+1)$ ) gives  $\mathcal{R}_{2,2} = 2^5 \cdot 5^4 \cdot 11^2 \cdot 17^3 \cdot 71$ . At  $p = 71$ :  $N_2(-5/2) \equiv N_4(-5/2) \equiv 0 \pmod{71}$  with  $-5/2 \notin \{-1, -2, -3, -4\}$ , a nonboundary common root inside the mesoscopic range ( $4^3 < 71$ ). This shows the boundary-only gcd statement is *false*.

**Remark 27** (Collision energy and the  $H^{1/2}$  barrier). The column exponent  $2/3$  is the natural barrier for the consecutive-gap argument: even if each gap of length  $d$  has only  $d$  (rather than  $3(d-1)$ ) admissible starts, the sum  $\sum_{d \leq L} d \asymp L^2$  still gives  $H/L + L^2$ , optimized at  $H^{2/3}$ . To achieve  $H^{1/2}$ , one needs the aggregate collision energy  $\mathcal{E}_p(H) = \sum_{d+r \leq H} \#\{x : N_d(x) \equiv N_r(x+d) \equiv 0\} \ll H^{3/2}$ , which is a sparse-exception theorem for the separated-block resultant family—beyond the scope of universal continuant identities. (The  $A_1$  T-system  $\tau_{h+1}(x)\tau_{h+1}(x+1) - \tau_{h+2}(x)\tau_h(x+1) = 1$  admits tame solutions with zero density  $1/2$ , e.g.  $\tau_{2k} = 0, \tau_{2k+1} = (-1)^k$ , so singularity confinement alone cannot force sparsity.) In the transfer-matrix formulation (writing  $M(n) = \begin{pmatrix} P(n) & -n^6 \\ 1 & 0 \end{pmatrix}$ , so that  $N_h(x) = 0$  iff a fixed matrix coefficient of the interval product  $M(x+h-1) \cdots M(x+1)$  vanishes), the conjectured  $\mathcal{E}_p(H) \ll H^{3/2}$  is analogous to a Rudnev-type point-plane incidence bound in  $\text{PGL}_2(\mathbb{F}_p)$ , subject to a nonconcentration hypothesis for the deterministic Apéry cocycle. Computation for selected primes:

| $p$                            | 71   | 503  | 2003 | 5003 | 7919 | 20011 | 50021 | $10^5+3$ |
|--------------------------------|------|------|------|------|------|-------|-------|----------|
| $H = \lfloor \sqrt{p} \rfloor$ | 8    | 22   | 44   | 70   | 88   | 141   | 223   | 316      |
| $R_p(H)/H$                     | 1.75 | 1.59 | 1.36 | 1.50 | 1.75 | 1.63  | 1.56  | 1.46     |
| $\mathcal{E}_p(H)$             | 1    | 2    | 2    | 2    | 10   | 0     | 2     | 0        |
| $\mathcal{E}_p(H)/H^{3/2}$     | .044 | .019 | .007 | .003 | .012 | 0     | .001  | 0        |

The ratio  $R_p(H)/H$  clusters around  $3/2$  (the Chebotarev prediction from Remark 17), and  $\mathcal{E}_p(H)/H^{3/2} \rightarrow 0$ , consistent with  $\mathcal{E}_p(H) \ll H^{3/2}$ . The key arithmetic object is the shifted resultant  $S_{d,r} = \text{Res}_x(N_d(x), N_r(x+d)) \in \mathbb{Z}$ : for  $p \nmid S_{d,r}$ , the gap polynomials  $N_d$  and  $N_r(\cdot+d)$  share no root over  $\mathbb{F}_p$ , so that gap pair contributes zero to  $\mathcal{E}_p$ . A Hadamard bound gives  $\log |S_{d,r}| \ll (d+r)^2 \log(d+r)$ , so the total number of prime divisors across all pairs  $(d, r)$  with  $d+r \leq H$  is  $O(H^4 \log H)$ . Averaging over the  $\sim X/\log X$  primes in  $(X, 2X]$  gives  $\mathcal{E}_p(H) \ll H^{1/2}$  for all but  $o(X/\log X)$  primes, provided  $H \leq X^{2/7-\varepsilon}$ —rigorous but far from the global target  $H = p-1$ .

A sharper route uses the bipartite incidence graph  $\Gamma_p(H)$ : left vertices are gap lengths  $d$ , right vertices are gap lengths  $r$ , with an edge when  $N_d$  and  $N_r(\cdot+d)$  share a root over  $\mathbb{F}_p$ . Each edge certifies a triple collision, so  $\mathcal{E}_p(H) = e(\Gamma_p(H))$ . A uniform fiber bound  $\deg_L(d) \leq C$  (the number of  $r$ -partners for each  $d$ ) and a uniform codegree bound (at most  $C$  common right neighbors for any two left vertices) would place  $\Gamma_p(H)$  in the regime of the Kővári–Sós–Turán theorem, giving  $\mathcal{E}_p(H) \ll H^{3/2}$ —the collision-energy scale needed for the mesoscopic target  $R_p(\sqrt{p}) = O(\sqrt{p})$ . The Dodgson identity (Remark 15) restricts local configurations in  $\Gamma_p$ , but by itself controls  $O(H)$  pairs out of  $O(H^2)$ ; upgrading it to a global codegree bound is the key missing step.

## 5.4 Computational evidence

**Proposition 28.** *For all 78,496 primes  $5 \leq p \leq 10^6$ :*

1.  $Z(p) \in \{0, 2, 4, 6, 8, 10, 12\}$  for all but two primes; the two exceptions ( $p = 11$  and  $p = 3137$ ) have  $Z(p) = 1$  (non-ordinary primes, where  $p \mid a_p(f)$ ).  $\max Z(p) = 12$  at  $p = 159,977$ .
2. The exact histogram is:

| $Z(p)$                    | 0     | 1    | 2     | 4    | 6    | 8    | 10   | 12   |
|---------------------------|-------|------|-------|------|------|------|------|------|
| count                     | 47632 | 2    | 23730 | 6045 | 951  | 123  | 12   | 1    |
| fraction                  | .607  | .000 | .302  | .077 | .012 | .002 | .000 | .000 |
| $\text{Poi}(\frac{1}{2})$ | .607  | —    | .303  | .076 | .013 | .002 | .000 | .000 |

The mean  $Z(p) = 1.000$ , stable across all ranges ( $\chi^2 = 3.42$  on 4 d.f. against  $\text{Poisson}(1/2)$  for the pair count). The empirical factorial moments of  $K(p) = Z(p)/2$  over the 78,494 ordinary primes are:

| $r$                                     | 1     | 2     | 3     | 4     | 5     | 6     |
|-----------------------------------------|-------|-------|-------|-------|-------|-------|
| $\frac{1}{\pi_{\text{ord}}} \sum (K)_r$ | .4998 | .2490 | .1210 | .0605 | .0275 | .0092 |
| $2^{-r}$                                | .5000 | .2500 | .1250 | .0625 | .0312 | .0156 |

The first four moments match  $\text{Poisson}(1/2)$  to better than 3%; the higher-moment deviation is consistent with sampling variance ( $K \geq 5$  has only 13 primes in this range).

3. The symmetry  $b_j \equiv b_{p-1-j} \pmod{p}$  holds for all tested primes: zeros come in palindromic pairs  $\{j, p-1-j\}$ , so  $Z(p)$  is even (except at non-ordinary primes). The pair count  $Z(p)/2$  fits  $\text{Poisson}(1/2)$  to within sampling noise (table above). Proposition 18 proves  $\text{Poisson}(1/2)$  rigorously in the prime-aspect; the orbit-aspect fit (Remark 19) remains empirical.



*Remark 29* (Mesoscopic root count). At the scale  $H = \lfloor \sqrt{p} \rfloor$ , the total root count  $R_p(H) = \sum_{h=2}^H r_p(h)$  satisfies  $R_p(H)/H \approx 3/2$  for all tested primes, consistent with the Chebotarev prediction from Remark 17:

| $p$    | $H = \lfloor \sqrt{p} \rfloor$ | $R_p(H)$ | $R_p(H)/H$ |
|--------|--------------------------------|----------|------------|
| 503    | 22                             | 35       | 1.59       |
| 2003   | 44                             | 60       | 1.36       |
| 5003   | 70                             | 105      | 1.50       |
| 10007  | 100                            | 154      | 1.54       |
| 20011  | 141                            | 230      | 1.63       |
| 50021  | 223                            | 347      | 1.56       |
| 100003 | 316                            | 460      | 1.46       |

The ratio  $R_p(H)/H$  stabilizes near  $3/2$  (range 1.32–1.75 for  $p > 500$ ), supporting the mesoscopic conjecture in Remark 23.

## 5.5 Hypothesis Z

**Hypothesis 30** (Hypothesis Z). *There exists an absolute constant  $C$  such that  $Z(p) \leq C$  for all primes  $p$ .*

*Remark 31.* The analogous problem for the  $\zeta(2)$  Apéry numbers  $A_n = \sum_k \binom{n}{k}^2 \binom{n+k}{k}$  has a fundamentally different structure. Stienstra–Beukers [20] showed that  $A_p$  is related to a CM modular form, yielding  $A_{(p-1)/2} \equiv 0 \pmod{p}$  for all  $p \equiv 3 \pmod{4}$  — a *positive density* of primes with guaranteed zeros. In contrast, the  $\zeta(3)$  Apéry numbers are governed by the non-CM newform  $f(z) = \eta(2z)^4 \eta(4z)^4 \in S_4(\Gamma_0(8))$  (LMFDB label 8.4.a.a), for which no such systematic zero-forcing occurs. This is a structural reason why Hypothesis  $\bar{Z}$  should hold for the  $\zeta(3)$  sequence, even though the pointwise bound  $Z(p) \leq C$  (Hypothesis Z) likely fails.

*Remark 32.* Even the single-row slice  $Z^*(p) := \mathbf{1}[p \mid b_{(p-1)/2}]$  is connected to the non-ordinary primes of  $f$  via the Beukers congruence  $b_{(p-1)/2} \equiv a_p(f) \pmod{p}$  [5, 1]. The density-zero conjecture for non-ordinary primes of non-CM weight-4 forms is itself open [14]. Thus Hypothesis Z contains a known open problem as a special case. The heuristic density of non-ordinary primes is  $\sim C_f/p$  (the Sato–Tate distribution on the normalized coefficients  $a_p/p^{3/2}$  admits  $\sim 4p^{1/2}$  multiples of  $p$  in  $[-2p^{3/2}, 2p^{3/2}]$ , each contributing mass  $\sim p^{-3/2}$ ). The expected count up to  $x$  is thus  $C_f \log \log x$  with  $C_f \approx 1$ . Finding 2 non-ordinary primes ( $p = 11, 3137$ ) below  $10^6$  is consistent with  $\log \log(10^6) \approx 2.9$ . This is not a standard Lang–Trotter problem: the target set  $\{a : p \mid a\}$  varies with  $p$ , and no analogue of Elkies’ theorem (infinitely many supersingular primes) is known for weight  $\geq 4$ .

*Remark 33* (Evidence against Hypothesis Z). The  $2 \cdot \text{Poisson}(1/2)$  distribution (Remark 7 below) has unbounded support: extreme-value theory predicts  $\max_{p \leq N} Z(p) \sim 2 \log \log N$ . The current record  $Z(p) = 12$  at  $p = 159,977$  (with record progression 2, 4, 6, 8, 10, 12) is consistent with this prediction. We therefore expect Hypothesis Z to fail: the first prime with  $Z(p) = 14$  should appear near  $p \approx 2 \times 10^7$  (the Gumbel prediction for the Poisson(1/2) pair maximum). A computation to  $p = 5 \times 10^7$  would either refute Hypothesis Z or provide strong evidence for a structural cutoff.

**Hypothesis 34** (Hypothesis  $\bar{Z}$ ).  $\sum_{p \leq x} Z(p) \ll \pi(x)$ , *equivalently, the average of  $Z(p)$  over primes  $p \leq x$  is bounded.*

*Remark 35.* Hypothesis  $\bar{Z}$  is strictly weaker than Hypothesis  $Z$  and consistent with  $Z(p) \sim 2 \cdot \text{Poisson}(1/2)$  (which gives  $\mathbb{E}[Z(p)] = 1$ , independent of  $p$ ). It suffices for the conditional bound  $\log G_n = O(\sqrt{n})$  (Theorem 55).

We state the full distributional prediction of the Poisson model (cf. Proposition 18):

**Hypothesis 36** (Poisson conjecture). *For every good prime  $p \geq 5$ , put*

$$\nu_p := \mathbf{1}_{\{p|a_p(f)\}}, \quad K_*(p) := \frac{Z(p) - \nu_p}{2},$$

*where  $f$  is the weight-4 newform associated to the Apéry differential operator. Thus  $K_*(p)$  counts noncentral reflected pairs of zeros of  $(b_j \bmod p)_{0 \leq j < p}$ . For every fixed  $r \geq 1$ ,*

$$\frac{1}{\pi(X)} \sum_{p \leq X} \binom{K_*(p)}{r} \rightarrow \frac{1}{2^r r!}.$$

*On ordinary primes ( $\nu_p = 0$ ), this reduces to  $K(p) = Z(p)/2 \rightarrow \text{Poisson}(1/2)$ , consistent with the data in Proposition 28.*

*Remark 37.* The Poisson conjecture implies Hypothesis  $\bar{Z}$ . Taking  $r = 1$  gives  $\sum_{p \leq X} K_*(p) = (\frac{1}{2} + o(1)) \pi(X)$ . Since  $Z(p) = 2K_*(p) + \nu_p$  and  $0 \leq \nu_p \leq 1$ ,

$$\sum_{p \leq X} Z(p) = 2 \sum_{p \leq X} K_*(p) + \sum_{p \leq X} \nu_p \leq (2 + o(1)) \pi(X),$$

which is Hypothesis  $\bar{Z}$ . A proof would require the fixed-order joint independence of reflection-pair zero events—the correlation theorem underlying the factorial-moment computation in Proposition 18—for the deterministic Apéry orbit, not just the hyperoctahedral random model.

*Remark 38* (Computational evidence for  $\bar{Z}$ ). Over dyadic blocks of primes,  $(\sum_{P < p \leq 2P} Z(p))/|\{P < p \leq 2P\}|$  ranges from 0.86 to 1.07 for  $P = 2^k$ ,  $3 \leq k \leq 12$  (666 primes up to 4999), with overall average  $\sum Z(p)/\pi = 0.91$ . The Poisson prediction is 1.

## 6 The denominator connection lemma

**Lemma 39.** *Let  $p \geq 7$  be prime with  $n/2 < p \leq n$ , and set  $r = n - p$ . Then  $v_p(G_n) = 0$  if and only if  $p \nmid b_r$ .*

*Proof.* Since  $p > n/2$ , the only multiple of  $p$  in  $\{1, \dots, n\}$  is  $p$  itself, so  $v_p(d_n) = 3$ .

**Step 1.**  $v_p(d_n b_n) = 3 + v_p(b_n) \geq 3$ , so  $v_p(G_n) = 0$  requires  $v_p(d_n a_n) = 0$ , i.e.,  $v_p(a_n) = -3$ .

**Step 2 (variation of parameters).** From Lemma 1, the Casorati identity telescopes to

$$\frac{a_n}{b_n} = \sum_{m=1}^n \frac{6}{m^3 b_m b_{m-1}}. \quad (12)$$

For  $p > n/2$ , the only  $m \in \{1, \dots, n\}$  with  $p \mid m$  is  $m = p$ .

**Step 3.** By Theorem 4,  $b_p \equiv b_0 b_1 = 5 \pmod{p}$  (since  $p$  has base- $p$  digits  $(0, 1)$ ), and by the symmetry  $b_j \equiv b_{p-1-j}$ ,  $b_{p-1} \equiv b_0 = 1 \pmod{p}$ . Since  $p \geq 7$ , both are nonzero mod  $p$ .

The  $m = p$  term in (12) has  $v_p = -3$ ; all other terms have  $v_p \geq 0$ . Therefore

$$p^3 \cdot \frac{a_n}{b_n} \equiv \frac{6}{b_p b_{p-1}} \equiv \frac{6}{5} \pmod{p}.$$

Using  $b_n \equiv 5b_r \pmod{p}$  (Theorem 4, since  $n = 1 \cdot p + r$ ):

$$p^3 a_n \equiv \frac{6}{5} \cdot 5b_r = 6b_r \pmod{p}.$$

Thus  $v_p(a_n) = -3$  iff  $p \nmid b_r$ . □

**Verification.** For all  $n \leq 300$  and primes  $p \in (n/2, n]$  with  $p \geq 7$ :  $v_p(G_n) = 0 \iff p \nmid b_{n-p}$  confirmed in 2101/2101 cases (100%).

**Corollary 40** (Lucas reduction). *For  $p \geq 7$  and  $p \in (n/2, n]$ :  $v_p(G_n) \geq 1$  iff  $p \mid b_{n-p}$ . By the Lucas congruence  $b_n \equiv b_1 b_{n-p} = 5b_{n-p} \pmod{p}$ , this is equivalent (for  $p \neq 5$ ) to  $p \mid b_n$ . Thus  $B(n) := \#\{p \in (n/2, n], p \text{ prime} : v_p(G_n) \geq 1\}$  equals  $\omega_{(n/2, n]}(b_n) + O(1)$ , the number of distinct prime factors of  $b_n$  lying between  $n/2$  and  $n$ , up to an absolute constant.*

The full conjecture  $G_n = e^{o(n)}$  for all  $n$  is therefore equivalent to the *top-half radical estimate*:

$$\log \text{rad}_{(n/2, n]}(b_n) = o(n). \quad (13)$$

Since  $\log b_n \sim n \log(17 + 12\sqrt{2}) \approx 3.52n$ , the height bound gives  $\omega_{(n/2, n]}(b_n) \leq \log b_n / \log(n/2) = O(n/\log n)$ , precisely one  $o(1)$  factor short of the target. Computation for  $n \leq 200,000$  gives  $B(n) \leq 3$  and  $B(n) = 0$  for 93.7% of values (see Remark 41 for the Poisson analysis).

*Remark 41* (Random-model prediction). Under a random-integer model,  $\Pr(p \mid b_n) \approx 1/p$  for each prime  $p$ , so by Mertens' theorem

$$\mathbb{E}[B(n)] = \sum_{n/2 < p \leq n} \frac{1}{p} = \log\left(\frac{\log n}{\log(n/2)}\right) + O(1/\log n) \approx \frac{\log 2}{\log n} \rightarrow 0.$$

Computation for  $n \leq 200,000$  confirms:  $\mathbb{E}[B] \approx 0.065$ ,  $\max B(n) = 3$ , and the histogram matches  $\text{Poisson}(\log 2 / \log n)$  to high accuracy:  $B(n) = 0$  for 93.7%,  $B(n) = 1$  for 6.0%,  $B(n) = 2$  for 0.20%,  $B(n) = 3$  for 0.006% (the Poisson predictions with  $\lambda = 0.065$  are 93.7%, 6.1%, 0.20%, 0.004%). The dispersion ratio  $\text{Var}(B)/\mathbb{E}[B] = 1.003$  is the exact Poisson signature. By CRT-based approximate independence across primes,  $B(n)$  should follow  $\text{Poisson}(\log 2 / \log n)$  with  $\lambda \rightarrow 0$ . Extreme-value theory then predicts  $\max_{n \leq N} B(n) = O(\log N / \log \log N)$ —trivially  $o(n/\log n)$ —so the random model *strongly* predicts the full conjecture. The obstruction is proving that  $b_n$  behaves “randomly” with respect to its prime factors in  $(n/2, n]$ , despite its highly structured factorization (supercongruences, Lucas decomposition, palindromic symmetry).

## 7 The unconditional density-one theorem

**Theorem 42.** *For a set of positive integers  $n$  of natural density 1,  $G_n = e^{o(n)}$ .*

*Proof.* Fix  $N$  large and consider  $n \in (N, 2N]$ . Write  $B(n) = \#\{\text{primes } p > \sqrt{n} \text{ with } v_p(G_n) \geq 1\}$ . By Proposition 3, the small-prime contribution is  $O(\sqrt{n})$  for every  $n$ . Each such prime contributes  $v_p(G_n) \log p \leq 3 \log n$  (since  $v_p(d_n) = 3$  for  $p > \sqrt{n}$ ), so  $\log G_n \leq O(\sqrt{n}) + 3B(n) \log n$ . It suffices to show  $B(n) = o(n/\log n)$  for all but  $o(N)$  values of  $n \in (N, 2N]$ .

For  $p > \sqrt{n}$ , write  $n = qp + r$  with  $1 \leq q \leq \sqrt{2N}$  and  $0 \leq r < p$ . By Lemma 47(iii),  $v_p(G_n) \geq 1$  requires  $D_q b_r \equiv 0 \pmod{p}$ . By Lemma 48, leading-digit bad primes ( $p \mid b_q$ ,  $p \nmid b_r$ ) have  $v_p(G_n) = 0$  and do not contribute to  $B(n)$ . The remaining contributions come from *lower-digit bad primes* ( $r \in \mathcal{Z}_p$ ) and *companion-block primes* ( $D_q \equiv 0 \pmod{p}$ ,  $p \nmid b_r$ ).

**Lower-digit incidence.** For fixed  $p \in (\sqrt{N}, 2N]$ , an interval of  $N$  consecutive integers contains at most  $N \cdot Z(p)/p + Z(p)$  values of  $n$  whose residue  $r = n \bmod p$  lies in  $\mathcal{Z}_p$ . Set  $\rho_p = Z(p)/p$ . The total lower-digit incidence is

$$R_N \leq \sum_{\sqrt{N} < p \leq 2N} (N\rho_p + Z(p)).$$

By Proposition 9,  $\rho_p \rightarrow 0$ . Since  $\pi(2N) = O(N/\log N)$ :

$$\sum_p \rho_p = o(\pi(2N)) = o(N/\log N), \quad \sum_p Z(p) = \sum_p p\rho_p \leq 2N \sum_p \rho_p = o(N^2/\log N).$$

Hence  $R_N = o(N^2/\log N)$ .

**Companion-block incidence.** By Proposition 51, the companion-block contribution to  $\log G_n$  is  $O(n^{2/3}) = o(n)$  at every index. Hence companion-block primes do not affect the density-one statement, and we set  $P_N = 0$  in the counting argument.

**Conclusion.** Write  $\sum_{n \in (N, 2N]} B(n) \leq R_N + P_N = o(N^2/\log N)$ . For any fixed  $\delta > 0$ , Markov's inequality gives

$$\#\{n \in (N, 2N] : B(n) > \delta n / \log n\} \leq \frac{o(N^2/\log N)}{\delta N / \log N} = o(N).$$

A dyadic decomposition shows the exceptional set has density zero, so  $B(n) = o(n/\log n)$  for density 1. For all such  $n$ ,  $\log G_n = O(\sqrt{n}) + 3B(n) \log n = O(\sqrt{n}) + o(n) = o(n)$ .  $\square$

**Corollary 43** (Quantitative exceptional set). *For every  $\varepsilon > 0$ ,*

$$\#\{n \leq N : \log G_n > \varepsilon n\} = O_\varepsilon(N^{2/3}).$$

*Proof.* We make the density-one argument quantitative using the explicit bound  $Z(p) = O(p^{2/3})$  from Proposition 9. For  $n \in (N, 2N]$ ,  $\log G_n \leq O(\sqrt{n}) + 3B(n) \log n$ , so  $\log G_n > \varepsilon n$  requires  $B(n) > \varepsilon n / (7 \log N)$  for  $N$  large.

Partial summation with  $Z(p) \leq Cp^{2/3}$  gives

$$\sum_{\sqrt{N} < p \leq 2N} \frac{Z(p)}{p} \leq C \sum_{p \leq 2N} p^{-1/3} = \frac{3}{2} C \frac{(2N)^{2/3}}{\log(2N)} + O(N^{1/3}) = O\left(\frac{N^{2/3}}{\log N}\right).$$

The lower-digit incidence satisfies  $R_N = N \sum Z(p)/p + \sum Z(p) = O(N^{5/3}/\log N)$  (the second sum,  $\sum Z(p) \leq C \sum p^{2/3}$ , contributes the same order). The companion-block incidence  $P_N = O(N^{3/2})$  is dominated by  $R_N$ . Hence  $\sum_{n \in (N, 2N]} B(n) = O(N^{5/3}/\log N)$ .

By Markov's inequality,

$$\#\{n \in (N, 2N] : B(n) > \varepsilon n / (7 \log N)\} \leq \frac{O(N^{5/3}/\log N)}{\varepsilon n / (7 \log N)} = O(N^{2/3}/\varepsilon).$$

A dyadic decomposition over intervals  $(N/2^{j+1}, N/2^j]$  gives the global bound, since  $\sum_{j \geq 0} (N/2^j)^{2/3} = O(N^{2/3})$ .  $\square$

In particular, the exceptional set has finite harmonic weight:

**Corollary 44** (Finite harmonic weight). *For every  $\varepsilon > 0$ ,*

$$\sum_{\substack{n \geq 1 \\ \log G_n > \varepsilon n}} \frac{1}{n} < \infty.$$

*Proof.* The contribution from  $(2^k, 2^{k+1}]$  is  $O(2^{-k/3})$  by Corollary 43, and the geometric series converges.  $\square$

*Remark 45.* The exponent  $2/3$  in the exceptional set is inherited from the lower-digit incidence  $R_N$ , which uses  $Z(p) = O(p^{2/3})$ . More generally, if  $Z_p(I) \leq C|I|^\alpha$  uniformly for all primes and sub-intervals (as Corollary 10 provides with  $\alpha = 2/3$ ), then  $R_N = O(N^{1+\alpha}/\log N)$  and  $L_N = O(N^{(4-\alpha)/(3-\alpha)}/\log N)$ . The exceptional set is therefore  $O(\max(N^\alpha, N^{1/(3-\alpha)})/\varepsilon)$ , with the crossover at  $\alpha = (3 - \sqrt{5})/2 \approx 0.382$ . Under Hypothesis  $\bar{Z}$ , the dyadic bound  $D(P, Q) \leq \sum Z(p) = O(P/\log P)$  is equivalent to  $\alpha = 0$  in this formula, giving the exceptional set  $O(N^{1/3}/\varepsilon)$  (Corollary 57). Theorem 55 shows the matching density-one bound  $\log G_n = O(\sqrt{n})$ , where  $O(\sqrt{n})$  is now the small-prime floor, not the leading-digit contribution.

## 8 Codegree amplification

We begin with a structural rigidity lemma: consecutive integers cannot share a top-half bad prime.

**Lemma 46** (Consecutive coprimality). *Let  $p > 5$  be prime and  $n \geq p$ . If  $p \mid b_{n \bmod p}$  and  $p \mid b_{(n+1) \bmod p}$ , then  $p \leq n/2$ . Equivalently, no prime  $p > n/2$  with  $p > 5$  can be lower-digit bad for both  $n$  and  $n + 1$ .*

*Proof.* Since  $p > n/2$ , both  $n$  and  $n + 1$  have base- $p$  representations  $n = p + r$ ,  $n + 1 = p + (r + 1)$  with  $0 \leq r < r + 1 < p$  (provided  $r < p - 1$ ; the case  $r = p - 1$  gives  $n + 1 = 2p$  and  $b_{n+1} \equiv b_2 b_0 = 73 \not\equiv 0$  for  $p > 73$ , a finite check).

Suppose  $p \mid b_r$  and  $p \mid b_{r+1}$  with  $r, r + 1 < p$ . The Apéry recurrence gives

$$(k + 1)^3 b_{k+1} = P(k) b_k - k^3 b_{k-1},$$

and since  $k + 1 < p$  implies  $(k + 1)^3$  is invertible mod  $p$ , the conditions  $p \mid b_k$  and  $p \mid b_{k+1}$  force  $p \mid b_{k-1}$ . Iterating downward from  $k = r$  gives  $p \mid b_j$  for all  $0 \leq j \leq r + 1$ , contradicting  $b_0 = 1$ .  $\square$

A prime that divides  $b_n$  only through a leading base- $p$  digit contributes nothing to  $G_n$ .

**Lemma 47** (Block system for  $p^3 a_n \bmod p$ ). *Let  $p \geq 7$ ,  $p \leq n < p^2$ , and write  $n = qp + r$  with  $1 \leq q < p$  and  $0 \leq r < p$ . Define  $\Delta_{q,r} \equiv p^3 a_{qp+r} \pmod{p}$  and  $D_q = \Delta_{q,0}$ . Then:*

- (i)  $\Delta_{q,r} \equiv D_q b_r \pmod{p}$  for every  $0 \leq r < p$ ;
- (ii)  $D_q b_{q-1} - D_{q-1} b_q \equiv 6/q^3 \pmod{p}$ ;
- (iii)  $p \mid G_n$  if and only if  $D_q b_r \equiv 0 \pmod{p}$ .

*Proof.* For  $p > \sqrt{n}$ ,  $v_p(\text{lcm}(1, \dots, n)) = 1$ , so  $v_p(d_n) = 3$ . Since  $d_n a_n \in \mathbf{Z}$ ,  $p^3 a_n \in \mathbf{Z}_p$ , and  $v_p(d_n a_n) = v_p(p^3 a_n)$ . Meanwhile  $v_p(d_n b_n) = 3 + v_p(b_n) \geq 3$ , so  $p \mid G_n \Leftrightarrow p \mid p^3 a_n$ .

(i) Multiply the Apéry recurrence at index  $qp + r$  by  $p^3$  and reduce modulo  $p$ . Since  $qp + r \equiv r \pmod{p}$ , the sequence  $r \mapsto \Delta_{q,r}$  satisfies the same recurrence  $(r + 1)^3 \Delta_{q,r+1} = P(r) \Delta_{q,r} - r^3 \Delta_{q,r-1}$

as  $r \mapsto D_q b_r$ , with matching initial values  $\Delta_{q,0} = D_q$  and  $\Delta_{q,1} = 5D_q = D_q b_1$ . Since  $(r+1)^3$  is invertible mod  $p$  for  $0 \leq r \leq p-2$ , uniqueness gives the identity.

(ii) Multiply the Wronskian  $a_n b_{n-1} - a_{n-1} b_n = 6/n^3$  at  $n = qp$  by  $p^3$  and reduce mod  $p$ , using  $b_{qp} \equiv b_q$  and  $b_{qp-1} \equiv b_{q-1} b_{p-1} \equiv b_{q-1}$  (Lucas), and  $p^3 a_{qp-1} \equiv D_{q-1} b_{p-1} \equiv D_{q-1}$  (part (i)).

(iii) Immediate from (i) and the observation above.  $\square$

**Lemma 48** (Leading-digit vanishing). *Let  $p \geq 7$ ,  $p \leq n < p^2$ ,  $n = qp + r$  with  $q \geq 1$ . If  $p \mid b_q$  and  $p \nmid b_r$ , then  $v_p(G_n) = 0$ .*

*Proof.* Since consecutive Apéry numbers below the  $p$ -th cannot both vanish mod  $p$  (the recurrence propagates to  $b_0 = 1$ ),  $p \nmid b_{q-1}$ . Part (ii) of Lemma 47 gives  $D_q b_{q-1} \equiv 6/q^3 \not\equiv 0 \pmod{p}$ , so  $D_q \not\equiv 0$ . By part (iii),  $p \mid G_n$  iff  $D_q b_r \equiv 0$ , but  $D_q \not\equiv 0$  and  $p \nmid b_r$ , so  $D_q b_r \not\equiv 0$  and  $v_p(G_n) = 0$ .  $\square$

*Remark 49* (Block constants are Apéry numbers). The block constant  $D_q$  of Lemma 47 equals  $a_q \pmod{p}$ , where  $a_q$  is the Apéry rational companion. Indeed,  $a_q$  satisfies the same Wronskian recurrence  $a_q b_{q-1} - a_{q-1} b_q = 6/q^3$  with  $a_0 = 0$ ,  $a_1 = 6$ , so  $D_q = a_q$  by uniqueness. Consequently,  $D_q \equiv 0 \pmod{p}$  iff  $p$  divides the numerator of  $a_q$  (in lowest terms) and  $p$  does not divide its denominator.

Since  $|a_q| \leq C(1 + \sqrt{2})^{4q}$  and  $\text{denom}(a_q) \mid \text{lcm}(1, \dots, q)^3$ , the numerator of  $a_q$  has  $O(q)$  prime divisors. In particular, for fixed  $q$ , at most  $O(q)$  primes  $p$  can be companion-block bad at quotient  $q$ . Computation through  $n = 2000$  confirms that the companion-block count  $\#\{p > \sqrt{n} : p \mid \text{num}(a_{\lfloor n/p \rfloor})\}$  is at most 3 for every  $n$  in this range.

**Proposition 50** (Reflection of the companion solution). *Let  $p \geq 5$  be prime. Every  $p$ -integral solution  $(u_n)$  of the Apéry recurrence (1) on  $\{0, \dots, p-1\}$  is palindromic modulo  $p$ :*

$$u_{p-1-j} \equiv u_j \pmod{p} \quad (0 \leq j \leq p-1).$$

*In particular, for the companion solution  $a_0 = 0$ ,  $a_1 = 6$ ,*

$$a_{p-1-j} \equiv a_j \pmod{p}, \quad a_{p-1} \equiv 0 \pmod{p}.$$

*Proof.* Let  $R$  denote the reversal operator,  $(Ru)_j = u_{p-1-j}$ . The coefficient polynomial  $P(n) = 34n^3 + 51n^2 + 27n + 5$  satisfies  $P(-1-x) = -P(x)$ , so substituting  $n \mapsto p-2-n$  in the recurrence shows that  $R$  maps  $p$ -integral solutions to  $p$ -integral solutions modulo  $p$ .

The integer Apéry solution  $(b_n)$  satisfies  $b_{p-1} \equiv 1$  and  $b_{p-2} \equiv 5 \pmod{p}$ : in the binomial sum  $b_n = \sum_k \binom{n}{k}^2 \binom{n+k}{k}^2$ , only the  $k=0$  term survives modulo  $p$  for  $n = p-1$ , while for  $n = p-2$  only  $k=0, 1$  survive, contributing  $1+4=5$ . Since  $Rb$  and  $b$  have the same initial values  $(b_0, b_1) = (1, 5)$ , uniqueness gives  $Rb = b$ .

For two solutions  $x, y$ , the discrete Wronskian  $W_n(x, y) = x_n y_{n-1} - x_{n-1} y_n$  satisfies  $(n+1)^3 W_{n+1} = n^3 W_n$ , so  $W_n(x, y) = C(x, y)/n^3$ . Using  $Rb = b$ :

$$W_n(Ru, b) = -W_{p-n}(u, b) = W_n(u, b).$$

Hence  $W_n(Ru - u, b) = 0$ , so  $Ru - u = cb$  for some  $c \in \mathbf{F}_p$ . Applying  $R$ : on one hand  $R(Ru - u) = -(Ru - u)$ , on the other  $R(cb) = cb$ . Since  $p$  is odd,  $c = 0$  and  $Ru = u$ .  $\square$

**Proposition 51** (Companion-block height bound). *Assume  $D_q \equiv a_q \pmod{p}$ . For every  $n \geq 1$ , the total logarithmic weight of companion-block primes is at most  $O(n^{2/3})$ :*

$$\sum_{\substack{p > \sqrt{n} \\ p \mid \text{num}(a_{\lfloor n/p \rfloor})}} \log p \leq C n^{2/3}$$

for an absolute constant  $C$ . In particular, companion-block primes contribute  $o(n)$  to  $\log G_n$  at every index.

*Proof.* Write  $a_q = A_q/C_q$  in lowest terms. From the telescoping identity  $a_q = b_q \sum_{k=1}^q 6/(k^3 b_k b_{k-1})$  and  $b_k \geq 1$ , we obtain  $|A_q| \leq |a_q| \cdot C_q \leq C_0(1 + \sqrt{2})^{4q}$ , hence each numerator  $A_q$  has at most  $O(q)$  prime divisors.

For a fixed  $n$ , set  $Q = \lfloor n^{1/3} \rfloor$ . A companion-block prime  $p > \sqrt{n}$  has quotient  $q = \lfloor n/p \rfloor < \sqrt{n}$ .

*Small quotients* ( $q \leq Q$ ): for each  $q$ , the primes dividing  $A_q$  contribute  $\sum \log p \leq \log |A_q| = O(q)$ . Summing over  $q \leq Q$ :  $\sum_{q=1}^Q O(q) = O(Q^2) = O(n^{2/3})$ .

*Large quotients* ( $q > Q$ ): each such prime satisfies  $p < n/Q = n^{2/3}$ . Since each prime determines its quotient uniquely, the total log-weight from this range is bounded by the Chebyshev sum  $\sum_{p < n^{2/3}} \log p = \theta(n^{2/3}) = O(n^{2/3})$ .  $\square$

The first-moment bound in Corollary 43 can be sharpened to a polylogarithmic exceptional set by exploiting the gap polynomials a second time, now as a *codegree* bound between nearby integers.

**Lemma 52** (Common bad-prime codegree). *Let  $m, n \in (N, 2N]$  with  $h = n - m \geq 1$ , and let  $P_0 \geq 1$ . The number of primes  $p > P_0$  that are simultaneously lower-digit bad for  $m$  and for  $n$  (i.e.,  $p \mid b_{m \bmod p}$  and  $p \mid b_{n \bmod p}$ ) is at most  $O(h \log N / \log P_0)$ . In particular, if  $P_0 \geq N^\delta$  for any fixed  $\delta > 0$ , the codegree is  $O(h)$ .*

*Proof.* Since  $p > N/2 \geq h$ , the residues  $m \bmod p$  and  $n \bmod p$  lie in  $\{0, \dots, p-1\}$  and differ by exactly  $h$ . By the gap implication (Lemma 7), if  $p \mid b_{m \bmod p}$  and  $p \mid b_{n \bmod p}$ , then  $p \mid N_h(m \bmod p)$ . The polynomial periodicity of  $N_h$  gives  $N_h(m \bmod p) \equiv N_h(m) \pmod{p}$ , so  $p \mid N_h(m)$ .

The gap polynomial has degree  $3(h-1)$  in  $m$  (the tridiagonal determinant representation gives exactly this), so for  $m \in (N, 2N]$ ,  $|N_h(m)| \leq (CN^3)^h$ . Since  $N_h(m) > 0$  (the associated tridiagonal matrix is positive definite at positive integers), the number of prime factors exceeding  $P_0$  is at most  $3h \log(CN) / \log P_0$ .  $\square$

**Theorem 53** (Polylogarithmic exceptional set). *For every  $\varepsilon > 0$ ,*

$$\#\{n \leq N : \log G_n > \varepsilon n\} = O_\varepsilon((\log N)^2).$$

*Proof.* Fix  $\varepsilon > 0$  and consider  $n \in (N, 2N]$  with  $\log G_n > \varepsilon n$ . By Proposition 3, the small-prime contribution is  $O(\sqrt{n})$ . Each prime  $p > \sqrt{n}$  satisfies  $v_p(G_n) \leq 3$  (since  $v_p(d_n) = 3$ ). By Lemma 48, leading-digit bad primes ( $p \mid b_{\lfloor n/p \rfloor}$  with  $n \bmod p \notin \mathcal{Z}_p$ ) satisfy  $v_p(G_n) = 0$ . The remaining primes with  $v_p(G_n) \geq 1$  are the lower-digit bad primes ( $n \bmod p \in \mathcal{Z}_p$ ) and the *companion-block primes* ( $D_q \equiv 0 \pmod{p}$  with  $p \nmid b_r$ , cf. Lemma 47). By Proposition 51, the total logarithmic weight of companion-block primes is  $O(n^{2/3})$  at every index, hence  $o(n)$ . Hence  $\log G_n \leq O(\sqrt{n}) + 3t(n) \log n$ , and  $\log G_n > \varepsilon n$  forces

$$t(n) := \#\{p > \sqrt{n} : n \bmod p \in \mathcal{Z}_p\} \geq \frac{\varepsilon n}{4 \log n} \quad (14)$$

for  $N$  sufficiently large.

Partition  $(N, 2N]$  into intervals  $J$  of length  $Y = c_1 \varepsilon^2 N / \log N$ , where  $c_1 > 0$  is a small constant to be chosen. Write  $\mathcal{E}_J = \{n \in J : \log G_n > \varepsilon n\}$  and  $M = |\mathcal{E}_J|$ .

For each prime  $p \in (\sqrt{N}, 2N]$ , let  $d_p$  count the elements of  $\mathcal{E}_J$  for which  $p$  is lower-digit bad. By (14),

$$I := \sum_p d_p \geq \frac{\varepsilon N M}{4 \log N}.$$

The prime universe has size  $L = \pi(2N) \leq 4N/\log N$ .

**Upper bound on pair incidence.** For  $m, n \in \mathcal{E}_J$  with  $m < n$ , Lemma 52 gives at most  $C(n - m)$  common lower-digit bad primes (taking  $P_0 = N^\delta$  in the codegree lemma). Hence

$$\sum_p \binom{d_p}{2} = \sum_{\substack{m < n \\ m, n \in \mathcal{E}_J}} |\mathcal{P}_m \cap \mathcal{P}_n| \leq C \sum_{\substack{m < n \\ m, n \in \mathcal{E}_J}} (n - m) \leq CYM^2.$$

**Lower bound on pair incidence.** By Cauchy–Schwarz applied to the  $d_p$ ,

$$\sum_p \binom{d_p}{2} \geq \frac{1}{2} \left( \frac{I^2}{L} - I \right) \geq \frac{1}{2} \left( \frac{\varepsilon^2 NM^2}{64 \log N} - I \right).$$

Combining gives  $CYM^2 \geq \varepsilon^2 NM^2/(128 \log N) - I/2$ . Since  $I \leq LM \leq 4NM/\log N$  and  $Y = c_1 \varepsilon^2 N/\log N$ , for  $c_1$  chosen smaller than  $1/(128C)$ , the  $YM^2$  term cannot absorb the quadratic lower bound once  $M > C'/\varepsilon^2$ . Hence  $M = O(1/\varepsilon^2)$ .

There are  $N/Y = O(\log N/\varepsilon^2)$  intervals. The total on  $(N, 2N]$  is therefore  $O(\log N/\varepsilon^4)$ . Summing over  $O(\log N)$  dyadic blocks  $(N/2^{j+1}, N/2^j]$  gives the result.  $\square$

Theorem 53 supersedes Corollary 43. The proof’s critical-window estimate—every interval of length  $\asymp \varepsilon^2 N/\log N$  in  $(N, 2N]$  contains  $O_\varepsilon(1)$  exceptional integers—directly gives the uniform local bound

$$\sup_{M \geq 0} \#\{M < n \leq M + L : \log G_n > \varepsilon n\} \ll_\varepsilon L^{2/3} + 1,$$

by applying the critical-window argument to any interval  $[M, M + L]$  with  $N = M$  and  $Y = c\varepsilon^2 M/\log M$ . Dividing by  $L$  and letting  $L \rightarrow \infty$  yields *upper Banach density zero* unconditionally, strengthening the harmonic-weight corollary 44.

## 9 Counting bad primes: the conditional result

We first sharpen the leading-digit incidence from  $O(N^{3/2})$  to  $O(N^{10/7}/\log N)$  via a dyadic argument combining divisor-counting with the sub-interval zero bound; this is unconditional and also strengthens the conditional theorem below.

**Proposition 54** (Dyadic leading-digit bound). *The leading-digit incidence  $L_N$  (counting pairs  $(n, p)$  with  $n \in (N, 2N]$ ,  $p > \sqrt{n}$  prime, and  $p \mid b_{\lfloor n/p \rfloor}$ ) satisfies*

$$L_N = O\left(\frac{N^{10/7}}{\log N}\right).$$

*Proof.* For dyadic blocks  $P = 2^j$ ,  $Q = 2^k$  with  $Q < P$  and  $PQ \in [N/4, 4N]$ , define

$$D(P, Q) = \#\{(p, q) : P < p \leq 2P, Q < q \leq 2Q, p \mid b_q\}.$$

Two bounds hold unconditionally:

- *Sub-interval zero counting:* By Corollary 10, each prime  $p \in (P, 2P]$  has at most  $O(Q^{2/3})$  zeros of  $b_j$  in  $(Q, 2Q]$ . Summing over  $\pi(2P)$  primes gives  $D(P, Q) = O(PQ^{2/3}/\log P)$ .



- *Divisor counting:* Since  $\log b_q \leq (q+1) \log 64$ , the number of prime factors of  $b_q$  exceeding  $P$  is at most  $\log b_q / \log P = O(q / \log P)$ . Summing over  $q \in (Q, 2Q]$  gives  $D(P, Q) \leq \sum_q O(q / \log P) = O(Q^2 / \log P)$ .

Hence  $D(P, Q) \leq \min(P Q^{2/3}, Q^2) / \log P$ .

Each pair  $(p, q)$  in  $D(P, Q)$  contributes to at most  $p \leq 2P$  values of  $n$  with  $\lfloor n/p \rfloor = q$ , so the block's contribution to  $L_N$  is at most  $D(P, Q) \cdot 2P \leq 2 \min(P^{4/3} N^{2/3}, N^2 / P) / \log P$  (using  $Q = \Theta(N/P)$ ). The crossover is at  $P_0 = N^{4/7}$ . Summing  $O(\log N)$  dyadic blocks  $P \in [\sqrt{N}, 2N]$ :

$$L_N \leq \sum_{P \leq N^{4/7}} \frac{2P^{4/3} N^{2/3}}{\log P} + \sum_{P > N^{4/7}} \frac{2N^2}{P \log P} = O\left(\frac{N^{10/7}}{\log N}\right),$$

since both sums are geometric series dominated by the  $P = N^{4/7}$  boundary term.  $\square$

**Theorem 55.** *Assume Hypothesis  $\bar{Z}$ . Then for a set of positive integers  $n$  of natural density 1,*

$$\log G_n = O(\sqrt{n}).$$

*Proof.* Fix  $N$  and  $\omega(N) \rightarrow \infty$  arbitrarily slowly. As in Theorem 42, decompose  $\log G_n = \sum_p v_p(G_n) \log p$ . The small-prime contribution is  $O(\sqrt{n})$ . Split  $B(n) = B_{\text{low}}(n) + B_{\text{lead}}(n)$ , where  $B_{\text{low}}$  counts bad primes  $p > \sqrt{n}$  with  $n \bmod p \in \mathcal{Z}_p$  and  $B_{\text{lead}}$  counts those with  $\lfloor n/p \rfloor \in \mathcal{Z}_p$ .

**Lower digit.** Under Hypothesis  $\bar{Z}$ , set  $M(t) = \sum_{p \leq t} Z(p) \ll t / \log t$ . Abel summation gives  $\sum_{\sqrt{N} < p \leq 2N} Z(p)/p = O(1)$ , so  $R_N = \sum B_{\text{low}}(n) = O(N)$ . By Markov,  $B_{\text{low}}(n) \leq \omega(N)$  for all but  $O(N/\omega(N)) = o(N)$  values.

**Leading digit.** As in the proof of Corollary 57, Hypothesis  $\bar{Z}$  gives  $L_N = O(N^{4/3} / \log N)$ . By Markov with threshold  $\omega(N) N^{1/3} / \log N$ :

$$\#\{n \in (N, 2N] : B_{\text{lead}}(n) > \omega(N) N^{1/3} / \log N\} \leq \frac{O(N^{4/3} / \log N)}{\omega(N) N^{1/3} / \log N} = O(N / \omega(N)) = o(N).$$

**Conclusion.** For all remaining  $n$ :

$$\log G_n \leq O(\sqrt{n}) + 3(\omega(N) + \omega(N) N^{1/3} / \log N) \log n.$$

Since  $\omega(N) \cdot N^{1/3} \ll N^{1/2}$  for any  $\omega(N) \rightarrow \infty$  slowly enough ( $\omega \leq N^{1/6}$  suffices), the second term is  $O(N^{1/2}) = O(\sqrt{n})$ .  $\square$

*Remark 56* (The leading-digit bottleneck). The gap between the conditional bound  $O(\sqrt{n})$  (Theorem 55) and the conjectured  $O(\log n)$  comes from the small-prime contribution (Proposition 3), which gives an unconditional floor of  $O(\sqrt{n})$ . Under Hypothesis  $\bar{Z}$ , the big-prime contribution is now absorbed into this floor, thanks to Proposition 54.

Heuristically,  $B_{\text{lead}}(n) = O(1)$  because the leading digit  $\lfloor n/p \rfloor$  lands in  $\mathcal{Z}_p$  with probability  $\leq C/p$  for each prime, giving expected  $B_{\text{lead}} = \sum C/p = O(1)$ . Making this rigorous would require  $D(P, Q) \ll Q / \log P$  (the random-divisibility prediction), improving the proven  $D(P, Q) \leq \min(P Q^{2/3}, Q^2) / \log P$ . The small-prime bound  $O(\sqrt{n})$  itself would need a finer analysis of  $v_p(G_n)$  for small primes.

**Corollary 57** (Conditional exceptional set). *Assume Hypothesis  $\bar{Z}$ . For every  $\varepsilon > 0$ ,*

$$\#\{n \leq N : \log G_n > \varepsilon n\} = O_\varepsilon(N^{1/3}).$$

*Proof.* Under Hypothesis  $\bar{Z}$ ,  $R_N = O(N)$ . For the leading-digit incidence, Hypothesis  $\bar{Z}$  also strengthens the dyadic bound: since each prime  $p \in (P, 2P]$  contributes at most  $Z(p)$  zeros to any sub-interval,  $D(P, Q) \leq \sum_{P < p \leq 2P} Z(p) = O(P/\log P)$  (using  $\sum_{p \leq x} Z(p) = O(\pi(x))$ ). The crossover with the divisor bound  $D(P, Q) \leq Q^2/\log P$  shifts from  $P_0 = N^{4/7}$  to  $P_0 = N^{2/3}$ , giving  $L_N = O(N^{4/3}/\log N)$ . The threshold  $B(n) > \varepsilon N/(7 \log N)$  is exceeded by at most

$$\frac{O(N) + O(N^{4/3}/\log N)}{\varepsilon N/(7 \log N)} = O(N^{1/3}/\varepsilon).$$

Dyadic summation over  $(N/2^{j+1}, N/2^j]$  gives the global bound.  $\square$

Under Hypothesis  $\bar{Z}$ , the exceptional set is not only sparse globally but cannot cluster in any interval:

**Corollary 58** (Upper Banach density zero). *Assume Hypothesis  $\bar{Z}$ . For every  $\varepsilon > 0$ , the set  $E_\varepsilon = \{n : \log G_n > \varepsilon n\}$  has upper Banach density zero:*

$$\lim_{L \rightarrow \infty} \sup_{M \geq 0} \frac{|E_\varepsilon \cap (M, M+L]|}{L} = 0.$$

*Proof.* For  $I = (M, M+L]$  with  $M > 2L$ , every prime  $p > n/2$  with  $n \in I$  lies in  $(M/2, M+L]$ . Under Hypothesis  $\bar{Z}$ , each such prime contributes at most  $Z(p)$  candidate bad integers to  $I$ , giving  $\sum_{n \in I} B(n) \leq \sum_{M/2 < p \leq M+L} Z(p) \ll \pi(M+L) = O(M/\log M)$ . For  $n \in I \cap E_\varepsilon$ , the bound  $\log G_n \leq O(\sqrt{n}) + 3B(n) \log n$  and  $\log G_n > \varepsilon n$  force  $B(n) \gg \varepsilon M/\log M$  for  $M$  large. Therefore  $|I \cap E_\varepsilon| \ll_\varepsilon 1 = o(L)$ . The case  $M \leq 2L$  follows similarly by covering  $(0, 3L]$  with  $O(\log L)$  dyadic intervals.  $\square$

**Corollary 59.** *The condition  $Z(p) = o(p)$  is necessary for the conjecture: if  $Z(p) \geq \varepsilon p$  for a positive proportion of primes, double-counting gives  $\log G_n \geq c\varepsilon n$  for a positive proportion of  $n$ . Conversely,  $Z(p) = o(p)$  implies  $G_n = e^{o(n)}$  for density 1 (Theorem 42).*

*Proof.* For each prime  $p \in (n/2, n]$  with  $Z(p) \geq \varepsilon p$ , Lemma 39 gives  $v_p(G_n) \geq 1$  whenever  $n-p \in \mathcal{Z}_p$ . Suppose a proportion  $\delta > 0$  of primes satisfy  $Z(p) \geq \varepsilon p$ . Summing over  $n \in (N, 2N]$  and bad primes  $p \in (n/2, n]$ , the total incidence count is  $\geq \delta \pi(2N) \cdot \varepsilon N/2 = \Omega(N^2/\log N)$ . Dividing by  $N$  and applying Markov's inequality: a proportion  $\geq \varepsilon\delta/4$  of integers  $n$  have  $B(n) \geq \varepsilon\delta N/(4 \log N)$ , giving  $\log G_n \geq 3B(n) \log(N/2) \geq c\varepsilon\delta n$ .  $\square$

*Remark 60* (The pointwise gap). The full conjecture  $G_n = e^{o(n)}$  for every  $n$  is strictly stronger than the density-one statement. By Corollary 40, it is equivalent to the top-half radical estimate (13). The height bound gives  $B(n) = O(n/\log n)$ , but for an *adversarial* family  $(Z_p)$  with  $|Z_p| \leq 1$ , one may choose  $Z_p = \{N-p\}$  for primes in  $(N/2, N]$ , producing  $B(N) \geq \pi(N) - \pi(N/2) \sim N/(2 \log N)$ . Thus the needed pointwise bound  $B(n) = o(n/\log n)$  requires arithmetic information specific to the Apéry sequence, not just the cardinalities  $Z(p)$ .

By Corollary 40, every bad prime  $p > n/2$  satisfies  $p \mid b_n$ . Hence the conjecture implies  $\log \text{rad}_{(n/2, n]}(b_n) = o(n)$ : the large prime radical of  $b_n$  must be subexponential, even though  $|b_n| = e^{\Theta(n)}$ .

The Poisson model predicts that  $B(n) \geq 1$  occurs infinitely often:  $\mathbb{P}(B(n) \geq 1) \sim \log 2/\log n$ , and  $\sum (\log n)^{-1}$  diverges. This is compatible with the conjecture, since one bad prime contributes only  $O(\log n)$  to  $\log G_n$ . To violate  $G_n = e^{o(n)}$ , one needs  $B(n) \geq \varepsilon n/(6 \log n)$ ; the Poisson tail gives  $\mathbb{P}(B(n) \geq k_n) \leq e^{-cn}$  for  $k_n = \lceil \varepsilon n/(6 \log n) \rceil$ , and the sum converges. Thus the model predicts: bad-prime occurrences are inevitable, but catastrophic pile-ups are finitely many.

## 10 Computational verification

We computed  $G_n$  exactly for  $n = 1, \dots, 1000$  using rational arithmetic.

| $n$ range | avg $\log(G_n)/n$ | max $\log(G_n)/n$ |
|-----------|-------------------|-------------------|
| 1–50      | 0.276             | 0.93              |
| 51–100    | 0.108             | 0.26              |
| 101–200   | 0.086             | 0.18              |
| 201–300   | 0.048             | 0.10              |
| 301–400   | 0.038             | 0.08              |
| 401–500   | 0.024             | 0.05              |
| 501–600   | 0.029             | 0.06              |
| 601–700   | 0.017             | 0.04              |
| 701–800   | 0.014             | 0.04              |
| 801–900   | 0.016             | 0.04              |
| 901–1000  | 0.013             | 0.03              |

A power-law fit gives  $\log(G_n)/n \sim 5n^{-0.87}$ , i.e.,  $\log G_n \sim 5n^{0.13}$ , far stronger than the density-one bound  $o(n)$  of Theorem 42.

The exponent 0.13 should be interpreted with caution: if  $\log G_n \sim \kappa \log n$ , the effective log-log slope is  $1/\log n = 0.145$  at  $n = 1000$ , close to the fitted 0.13. A random-gcd model—assigning each prime  $p \leq n$  an independent probability  $\sim 1/p$  of dividing  $G_n$ —predicts  $\mathbb{E}[\log G_n] \sim \log n$  by the weighted Mertens theorem, with standard deviation  $\sim (\log n)/\sqrt{2}$  of the same order. The data thus cannot yet distinguish  $\kappa \log n$  from  $n^{0.13}$ ; the logarithmic prediction is arithmetically more natural.

## 11 Remarks

1. The Beukers supercongruence  $b_{mp^r} \equiv b_{mp^{r-1}} \pmod{p^{3r}}$  [4] provides tower rigidity: for each fixed prime,  $v_p(b_n)$  is controlled by the base- $p$  digits. Quantitatively, since  $b_n \equiv \prod b_{d_i} \pmod{p}$  (Theorem 4), we have  $p \nmid b_n$  iff every base- $p$  digit of  $n$  avoids  $\mathcal{Z}_p$ . Hence

$$\#\{n < p^r : p \mid b_n\} = p^r - (p - Z(p))^r.$$

2. For the gap-2 polynomial  $C_2(m) = P(m+1)$ , the factorization  $P(x) = (2x+1)(17x^2+17x+5)$  shows that gap-2 pairs can only occur at  $m = (p-3)/2$  (when  $2(m+1)+1 = 2m+3 \equiv 0$ ) or at roots of the quadratic  $17x^2+17x+5$  (discriminant  $-51$ ; exists over  $\mathbb{F}_p$  iff  $(\frac{-51}{p}) = 1$ ). Empirically, all observed gap-2 pairs arise from the linear factor at  $m = (p-3)/2$ .
3. For  $1 \leq j \leq p-2$ , the Apéry number  $b_j \bmod p$  admits a character-sum representation as a Greene [12] Gaussian hypergeometric  ${}_4F_3$  over  $\mathbb{F}_p$  with parameter  $\chi = \omega^j$  (Teichmüller character). However, the Weil bound controls the archimedean size  $|b_j|_\infty = O(p^{3/2})$  but does *not* bound  $Z(p)$ , since zeros of  $b_j$  modulo  $p$  are a  $p$ -adic event. The gap polynomial argument above is specifically adapted to this  $p$ -adic zero-count problem.
4. Malik–Straub [15] studied Lucas congruences and zero-divisibility for all sporadic Apéry-like sequences; Rowland–Yassawi [18] proved automaticity of the Lucas property for diagonals of rational functions, which includes the Apéry sequence. Our  $Z(p)$  data for the  $\zeta(3)$  sequence agrees with their empirical findings.

5. In the standard toric coordinate, the truncated polynomial  $H_p(t) = \sum_{j=0}^{p-1} b_j t^j \in \mathbb{F}_p[t]$  is the scalar Hasse–Witt invariant of the Apéry K3 pencil. Its *roots* in  $\mathbb{F}_p$  are the nonordinary fibers; because the family has generic Picard rank 19, nonordinary implies supersingular at good primes. Our statistic  $Z(p)$  counts vanishing *coefficients* of  $H_p$ , equivalently the vanishing finite Mellin coefficients  $b_j = -\sum_{t \in \mathbb{F}_p^\times} H_p(t) t^{-j}$  for  $1 \leq j \leq p-2$ . Thus  $Z(p)$  is a spectral sparsity statistic for the global Hasse/point-count function, *not* a count of supersingular fibers. Standard Hasse-stratification, Newton-polygon, and overconvergent  $F$ -crystal theory do not by themselves bound  $Z(p)$ ; they control the root locus, not the Fourier support. A bounded-conductor realization of  $H_p$  as an  $\ell$ -adic trace function could connect to the Forey–Fresán–Kowalski framework [11], but equidistribution of complex traces does not directly control exact zeros modulo  $p$ .
6. **Palindromic symmetry.** For  $0 \leq j \leq p-1$  and  $0 \leq k \leq p-1-j$ , the identity  $\binom{p-1-j}{k} \equiv (-1)^k \binom{j+k}{k} \pmod{p}$  holds (each of the  $k$  falling factors  $p-1-j-i \equiv -(j+1+i)$ ). When  $k \leq j$ ,  $\binom{p-1-j+k}{k} \equiv (-1)^k \binom{j}{k}$ ; when  $k > j$ , Lucas’s theorem gives  $\binom{p-1-j+k}{k} \equiv 0$  (since  $p-1-j+k \geq p$  and  $\binom{k-j-1}{k} = 0$ ). Therefore each term of  $b_{p-1-j} = \sum_k \binom{p-1-j}{k}^2 \binom{p-1-j+k}{k}^2$  with  $k > j$  vanishes, and for  $k \leq j$  the  $(-1)^{2k}$  signs cancel, giving  $b_{p-1-j} \equiv b_j \pmod{p}$ . The extra terms of  $b_j$  when  $j > (p-1)/2$  (i.e.,  $k > p-1-j$ ) likewise vanish:  $\binom{j+k}{k} \equiv \binom{j+k-p}{k} \equiv 0$  since  $j+k-p < k$ . Equivalently,  $H_p(t)$  is palindromic:  $t^{p-1} H_p(1/t) \equiv H_p(t)$ . This explains the observed pairing: whenever  $b_j \equiv 0$ , also  $b_{p-1-j} \equiv 0$ , so zeros of  $b_j$  come in pairs  $\{j, p-1-j\}$  (except at the self-paired center  $j = (p-1)/2$ , which vanishes only at non-ordinary primes). Proposition 50 proves the stronger statement that *every* solution of the Apéry recurrence is palindromic modulo  $p$ , in particular the companion solution  $a$ . The “pair count”  $K(p) = Z(p)/2$  (or  $(Z(p)-1)/2$  at non-ordinary primes) follows Poisson(1/2) to the precision of the data (Proposition 28).
7. **Sym<sup>2</sup> squareness (theorem).** The Apéry differential operator is the symmetric square of the Picard–Fuchs operator of an underlying elliptic family (Stienstra–Beukers [20]). Caruso, Fürsinn, Vargas-Montoya, and Zudilin [7] proved the mod- $p$  squareness rigorously: for every prime  $p \geq 5$ , setting  $\varepsilon_p = \frac{1}{2}(1 - (\frac{-6}{p}))$ , there exists a unique (up to sign) polynomial  $B_p \in \mathbb{F}_p[t]$  such that

$$H_p(t) = \Delta(t)^{\varepsilon_p} \cdot B_p(t)^2,$$

where  $\Delta(t) = t^2 - 34t + 1$  is the conifold discriminant and  $B_p$  is squarefree and coprime to  $\Delta$ . The proof uses the explicit identity  $F(t(x)) = (1+x)h(x)^2$  between the Apéry and Franel generating functions, the  $p$ -Lucas property of both sequences, and a quadratic-descent argument through the covering  $\mathbb{F}_p(x)/\mathbb{F}_p(t)$  where  $t = x(1-8x)/(1+x)$ . The deck-transformation character  $(-6/p)$  determines whether  $\varepsilon_p = 0$  or 1:

$$\varepsilon_p = 0 \iff \left(\frac{-6}{p}\right) = 1, \quad \varepsilon_p = 1 \iff \left(\frac{-6}{p}\right) = -1.$$

Equivalently,  $\varepsilon_p = 0$  iff  $p \equiv 1, 5, 7, 11 \pmod{24}$ . These two cases occur with equal density 1/2 by Dirichlet’s theorem.

Splitting the mean  $Z(p)$  by residue class modulo 8 (and modulo 24) yields no significant difference (mean  $\approx 1.00$  in each class,  $n \geq 2000$  primes per class up to  $10^5$ ); the squareness structure does not directly influence  $Z(p)$ , consistent with  $Z(p)$  counting coefficient zeros rather than root zeros.

The factorization imposes a convolution structure:  $b_j$  is (up to the  $\Delta^{\varepsilon_p}$  correction) the  $j$ -th coefficient of  $B_p^2$ , i.e.,  $b_j \equiv \sum_k \beta_k \beta_{j-k} \pmod{p}$  where  $\beta_k$  denotes the coefficients of  $B_p$ . The vanishing  $b_j \equiv 0$  thus requires cancellation across  $\sim p/2$  convolution terms. A Monte Carlo simulation (1000 random monic polynomials of degree  $(p-1)/2$  over  $\mathbb{F}_p$ , for  $p \in \{101, 503, 1009, 5003\}$ ) shows that the zero-count of the *squared* polynomial follows Poisson(1) (mean  $\approx 1$ , variance  $\approx 1$ , total variation distance to Poisson(1) equal to 0.018). However, the observed  $Z(p)$  distribution over 667 primes  $p \leq 5000$  is *not* Poisson(1) (TV = 0.43); instead it matches  $2 \cdot \text{Poisson}(1/2)$  with TV = 0.023. The explanation is the palindromic pairing: since  $b_{p-1-j} \equiv b_j$ , zeros come in pairs  $\{j, p-1-j\}$  and  $Z(p)$  is almost always even. Denoting by  $Z_{\text{pair}}(p)$  the number of zero pairs, the data show  $Z_{\text{pair}} \sim \text{Poisson}(1/2)$ , so  $Z(p) = 2Z_{\text{pair}}(p) \sim 2 \cdot \text{Poisson}(1/2)$ . Thus squareness explains the  $O(1)$  mean, while palindromic symmetry supplies the parity constraint. The Poisson tail predicts  $\max_{p \leq N} Z(p) \sim 2 \log \log N$  via extreme-value theory, strongly suggesting that  $Z(p)$  is unbounded (i.e., Hypothesis Z is false). Whether the algebraic provenance of  $B_p$  imposes a bounded *average* (Hypothesis  $\bar{Z}$ ) remains the central open question.

The factorization does *not* improve the divisor-counting bound  $D(P, Q) \leq Q^2/\log P$ . In the regime  $P \geq Q^{4/3}$  where the divisor estimate is active, one has  $q < p/2$ , so the coefficient  $b_q \pmod{p}$  lies in the formally free triangular range of the square root: the map  $(\beta_1, \dots, \beta_q) \mapsto (b_1, \dots, b_q) \pmod{p}$  is a triangular automorphism with Jacobian  $2^q$ , hence the single equation  $b_q \equiv 0$  imposes no algebraic constraint beyond fixing one coefficient of  $B_p$ . Thus the  $\text{Sym}^2$  structure gives the correct *random model* (uniform square root  $\Rightarrow$  Poisson(1/2)) but not the missing *distribution theorem* for the deterministic Apéry square root.

8. **Toric framework.** The Laurent polynomial  $f = w(x + x^{-1} + y + y^{-1} + z + z^{-1})$  on  $(\mathbb{F}_p^\times)^4$  has Newton polytope  $\Delta = \text{conv}(\{0\} \cup \text{Supp}(f))$ , a 4-dimensional pyramid over the regular octahedron in  $\mathbb{R}^3$ . Its  $f$ -vector is  $(7, 18, 20, 9)$ , its normalized volume is  $4! \cdot \text{vol}(\Delta) = 8$ , and it is Kouchnirenko nondegenerate for every prime  $p \geq 5$ : for each of the 27 nonempty faces  $\sigma$  not containing the origin, the toric critical system  $\{x_i \partial f_\sigma / \partial x_i = 0\}$  has no solution in  $(\mathbb{F}_p^\times)^4$ . For the Adolphson–Sperber twisted Hodge polygon with  $w$ -Kummer twist  $\chi_w = \omega^{-j}$  (character parameter  $a = j/(p-1)$ ), the slopes are  $a, a+1, a+2, a+3$  with multiplicities  $1, 3, 3, 1$ , confirming that the first slope equals  $a$  for every  $a \in \{k/100 : 0 \leq k \leq 99\}$ . By Newton-above-Hodge (Adolphson–Sperber [2]), the  $p$ -adic Newton polygon lies on or above this Hodge polygon, and ordinarity (Newton = Hodge) implies the existence of a unit root. The  $Z(p)$  bound thus reduces to proving ordinarity for all but  $O(1)$  twist parameters  $j$ .
9. **Branch obstruction.** The  $p-1$  Kummer twists  $\chi^{-j}$  ( $j = 0, \dots, p-2$ ) are distinct characters of conductor  $p$ . In Iwasawa-theoretic coordinates  $\chi^j = \omega^a \langle \cdot \rangle^s$  with  $a = j \pmod{p-1}$ , the horizontal family  $\{j : 0 \leq j < p-1\}$  hits *each* of the  $p-1$  branches  $\omega^a$  exactly once. Weierstrass preparation and bounded- $\lambda$  rigidity control zeros *within one branch* (the vertical direction  $j, j + (p-1), j + (p-1)p, \dots$ , where Dwork congruences live) but not *across* branches. Thus there is no  $p$ -adic analytic function of one variable whose zero count majorizes  $Z(p)$ . Any “rigidity” proof of Hypothesis  $\bar{Z}$  would need a mechanism not supplied by existing Iwasawa–Dwork theory.
10. Any estimate  $\log G_n = o(n)$ —whether  $O(\log n)$ ,  $O(\sqrt{n})$ , or  $O(n^{0.13})$ —yields  $\gamma = 0$  in the exponent  $G_n = e^{\gamma n + o(n)}$  and therefore does not improve the Rhin–Viola irrationality measure  $\mu(\zeta(3)) \leq 5.513$  [19], which depends only on the exponential growth rate  $|b_n \zeta(3) - a_n| = e^{-(\log \alpha + o(1))n}$ . An improvement would require an exponentially large common divisor, which is not predicted by any current model.

11. **Comparison with previous work.** Classical refinements of Apéry’s method (Beukers [4], Rhin–Viola [19], Zudilin [21]) exhibit explicit common factors of the hypergeometric coefficients to improve denominator estimates and irrationality measures, but always uniformly for every index. They do not provide exceptional-set estimates for the *actual* reduced gcd  $G_n$ . A separate body of work, based on the Subspace Theorem, proves  $\gcd(a^n - 1, b^n - 1) = e^{o(n)}$  pointwise for multiplicatively independent  $a, b$  (Bugeaud–Corvaja–Zannier [6]); these results apply to constant-coefficient linear recurrences via Binet representations, but the Apéry recurrence’s polynomially varying coefficients preclude the multiplicative-independence hypothesis. To our knowledge, no previous work gives a quantitative power-saving exceptional-set estimate for gcd cancellation in a variable-coefficient holonomic recurrence.
12. **The remaining open problem.** The companion-block height bound (Proposition 51) shows that primes  $p > \sqrt{n}$  with  $D_q \equiv 0 \pmod{p}$  contribute  $O(n^{2/3})$  to  $\log G_n$  at *every* index. After removing this channel, the conjecture  $G_n = e^{o(n)}$  for all  $n$  is equivalent to the *pointwise radical estimate*

$$\sum_{\substack{p > \sqrt{n} \\ p | b_n \bmod p}} \log p = o(n).$$

Theorem 53 proves this for all but  $O_\varepsilon((\log N)^2)$  integers in  $[1, N]$ . However, the codegree method (Lemma 52) controls primes *shared* between pairs of integers, and does not bound the total weight of “private” primes at a single index. Closing the gap requires a one-point rainbow estimate: an argument that one fixed integer cannot simultaneously be bad at too many distinct primes.

## 12 The second-moment approach

The codegree method of Section 8 exploits primes shared between pairs of integers to prove a density-one theorem. To close the remaining gap—proving  $W(n) = o(n)$  for *every*  $n$ —we study the *second moment* of the bad-prime weight function.

### 12.1 The criterion

For a dyadic shell  $I_N = (N, 2N]$ , define

$$B(m) = \sum_{\substack{p > \sqrt{m} \\ p | b_m \bmod p}} \log p, \quad M_2(N) = \sum_{m \in I_N} B(m)^2.$$

**Proposition 61** (Second-moment criterion). *If  $M_2(N) \leq C \cdot N^{2-\delta}$  for some  $C > 0$  and  $\delta > 0$ , then  $W(n) = o(n)$  for all  $n$ . More precisely, for every  $n \in I_N$ ,*

$$W(n) \leq B(n) \leq \sqrt{M_2(N)} \leq C^{1/2} \cdot N^{1-\delta/2}.$$

*Proof.*  $B(n)^2$  is one nonneg. term of the sum  $M_2(N)$ , hence  $B(n) \leq \sqrt{M_2(N)}$ . □

In particular,  $M_2(N) = O(N \cdot (\log N)^A)$  for any fixed  $A$  gives  $B(n) = O(\sqrt{N} \cdot (\log N)^{A/2}) = o(N)$ .

*Remark 62* (Shifted-gcd identity). The bad-prime weight admits a combinatorial reformulation. By the injectivity of  $p \mapsto n \bmod p$  on primes  $p > \sqrt{n}$  (two such primes giving the same residue would force their product  $\leq n$ ),

$$W(n) = \sum_{0 < r < n} \log \text{rad}_{>\max(\sqrt{n}, r)} \gcd(n-r, b_r).$$

Each inner radical is 1 or a single prime  $p = n - r$ . For the top-half block  $p > n/2$ , the condition becomes “ $n - r$  is prime and  $(n - r) \mid b_r$ .”

**Proposition 63** (Hard-core reduction). *For each bad prime  $p > \sqrt{n}$ , write  $n = q_p p + r_p$ ,  $s_p = \min(r_p, p-1-r_p)$ ,  $h_p = |p-1-2r_p|$ . Let  $1 \leq Q, R, H < n/2$ . Then the total weight of primes satisfying at least one of  $q_p > Q$ ,  $s_p \leq R$ ,  $h_p \leq H$  is  $O(n/Q + (R+H)\log n)$ . In particular, with  $Q = R = H = \sqrt{n/\log n}$ , all “easy channels” contribute  $O(\sqrt{n \log n})$ ; the remaining hard core consists of primes  $p \gg \sqrt{n \log n}$  with zero positions far from both endpoints and the reflection centre.*

*Proof.* Large quotient ( $q_p > Q$ ): then  $p < n/Q$ , so  $\sum \log p \leq \vartheta(n/Q) \ll n/Q$ . Near endpoint ( $s_p \leq R$ ): by the injectivity of  $p \mapsto n \bmod p$ , at most two primes  $> \sqrt{n}$  contribute for each value of  $s_p$ . Near centre ( $h_p \leq H$ ): the identity  $(2q_p + 1)p = 2n + 1 \pm h_p$  forces  $p$  to divide a fixed integer of size  $O(n)$ , giving  $O(1)$  primes per value of  $h_p$ .  $\square$

*Remark 64* (The vertical main term is harmless). Write  $W(n) = M(n) + \mathcal{E}(n)$  where  $M(n) = \sum_{p > \sqrt{n}} Z(p) (\log p)/p$ . The unconditional bound  $Z(p) \leq Cp^{2/3}$  gives  $M(n) \ll \sum_{p \leq n} p^{-1/3} \log p \ll n^{2/3}$ . The entire problem is the oscillatory error  $\mathcal{E}(n)$ , which at each prime is  $(\log p) \cdot (1_{n \bmod p \in \mathcal{Z}_p} - Z(p)/p)$ .

## 12.2 Empirical verification

The following table, computed for all dyadic shells with  $64 \leq N \leq 4096$  using the full zero-set data for all primes  $p \leq 20,000$ , gives:

| $N$  | $M_2(N)$ | $M_2/N$ | $M_2/(N \log^2 N)$ | $\max B(m)$ |
|------|----------|---------|--------------------|-------------|
| 64   | 672      | 10.5    | 0.605              | 10.4        |
| 128  | 1,684    | 13.2    | 0.560              | 15.9        |
| 256  | 4,376    | 17.1    | 0.555              | 15.8        |
| 512  | 11,896   | 23.2    | 0.598              | 22.3        |
| 1024 | 28,632   | 28.0    | 0.583              | 21.7        |
| 2048 | 69,226   | 33.8    | 0.582              | 27.1        |
| 4096 | 166,671  | 40.7    | 0.587              | 32.1        |
| 8192 | 421,868  | 51.5    | 0.634              | 37.2        |

The ratio  $M_2(N)/(N \log^2 N)$  stabilizes near 0.59, strongly suggesting

$$M_2(N) \sim C_2 \cdot N \cdot (\log N)^2, \quad C_2 \approx 0.59. \quad (15)$$

This gives  $B(n) \leq \sqrt{0.59 N} \log N \approx 0.77 \sqrt{N} \log N = o(N)$ .

### 12.3 CRT expansion

Expand  $M_2$  dyadically. For distinct primes  $p, q \in (P, 2P]$  with  $P > \sqrt{N}$ , write  $\Omega_p = \{m \in I_N : m \bmod p \in \mathcal{Z}_p\}$ . Then

$$M_2(N) = \underbrace{\sum_p (\log p)^2 |\Omega_p|}_{\text{diagonal}} + \underbrace{\sum_{p \neq q} \log p \log q |\Omega_p \cap \Omega_q|}_{\text{off-diagonal}}.$$

**Diagonal.**  $|\Omega_p| = Z(p) \cdot N/p + O(Z(p))$ , so

$$\text{diag} = N \sum_p \frac{Z(p) (\log p)^2}{p} + O\left(\sum_p Z(p) (\log p)^2\right).$$

Under the Poisson model  $\mathbb{E}[Z(p)] = 1$ , the main term is  $\sim N \cdot \sum_{P < p \leq 2P} (\log p)^2 / p \sim N \cdot (\log P)^2 \cdot \log 2$ . Summing dyadically  $P = \sqrt{N}, 2\sqrt{N}, \dots, N$  gives diagonal  $\sim \frac{1}{2} N (\log N)^2$ , matching  $\approx 63\%$  of the empirical (15).

**Off-diagonal.** Since  $p, q > \sqrt{N}$  we have  $pq > N$ , so each CRT class  $r_p \pmod{p}$ ,  $r_q \pmod{q}$  contains at most one element of  $I_N$ . The CRT main term of the off-diagonal is  $\sim N (\sum Z(p) \log p / p)^2 \sim \frac{1}{4} N (\log N)^2$ .

**The predicted constant.** Under the independent random model—each prime  $p$  hits a uniform  $m$  with probability  $\sim 1/p$ , independently across primes—the second moment of  $B(m)$  at one integer is  $\mathbb{E}[B(m)^2] = (\mathbb{E}[B])^2 + \text{Var}(B)$ . By weighted Mertens,  $\mathbb{E}[B] = \sum_{\sqrt{m} < p \leq m} (\log p) / p = \frac{1}{2} \log m + O(1)$ , and

$$\text{Var}(B) = \sum_{\sqrt{m} < p \leq m} \frac{(\log p)^2}{p} \left(1 - \frac{1}{p}\right) = \frac{3}{8} (\log m)^2 + O(\log m).$$

Therefore

$$\mathbb{E}[B(m)^2] = \left(\frac{1}{4} + \frac{3}{8}\right) (\log m)^2 + O(\log m) = \frac{5}{8} (\log m)^2 + O(\log m), \quad (16)$$

predicting  $M_2(N) \sim \frac{5}{8} N (\log N)^2$ . The diagonal constant is  $3/8$  (matching the observed  $\text{Diag}/N \log^2 N \approx 0.37$ ) and the off-diagonal contributes  $1/4$ . The empirical ratio  $M_2/(N \log^2 N) \approx 0.59$  is consistent with convergence to  $5/8 = 0.625$ , with lower-order boundary effects.

**Centered variance and cross-prime covariance.** The centered dispersion  $V^\circ = M_2 - M_1^2/N$  decomposes as

$$V^\circ = \underbrace{\sum_p (\log p)^2 |\Omega_p| (1 - |\Omega_p|/N)}_{\leq \text{Diag}} + \underbrace{\sum_{p \neq q} \log p \log q (|\Omega_p \cap \Omega_q| - |\Omega_p| |\Omega_q|/N)}_{E_N^\circ}. \quad (17)$$

The first sum is at most the diagonal  $S$  (since  $|\Omega_p|/N \geq 0$ ). The term  $E_N^\circ$  is the centered cross-prime covariance: it vanishes exactly when the events  $\{m \in \Omega_p\}$  are pairwise uncorrelated. The computed values are:



| $N$   | $M_2/N\log^2 N$ | $\text{Diag}/N\log^2 N$ | $V^\circ/S$ | $E_N^\circ$ |
|-------|-----------------|-------------------------|-------------|-------------|
| 256   | 0.556           | 0.359                   | 1.014       | +110        |
| 512   | 0.597           | 0.376                   | 1.010       | +215        |
| 1024  | 0.582           | 0.375                   | 0.981       | -152        |
| 2048  | 0.581           | 0.370                   | 0.992       | +7          |
| 4096  | 0.588           | 0.371                   | 0.995       | +41         |
| 8192  | 0.634           | 0.387                   | 0.995       | -188        |
| 16384 | 0.669           | 0.400                   | 0.995       | -922        |
| 32768 | 0.646           | 0.394                   | 0.988       | -14065      |

The ratio  $V^\circ/S$  converges to 1 from below for  $N \geq 1024$ , with  $V^\circ/S < 1$  at every scale tested beyond 512. The negative sign of  $E_N^\circ$  for large  $N$  reflects genuine *anti-correlation* among the CRT-constrained incidence events: at  $N = 1024$ , over 96% of cross-prime covariance pairs  $C_{p,q}$  are negative. This is consistent with a weak form of negative dependence arising from the partial-period CRT constraint. The magnitude  $|E_N^\circ|$  is  $O(\sqrt{N}(\log N)^2)$ , so  $|E_N^\circ|/S \rightarrow 0$  at rate  $\sim 1/\sqrt{N}$ .

**Three-model comparison.** To test whether the observed  $V^\circ/S \approx 1$  is an Apéry-specific phenomenon or a generic consequence of the cardinality and reflection structure, we compare three models on each dyadic block  $I_N$ : (1) the actual Apéry zero-sets; (2) uniformly random reflected subsets of  $\mathbb{F}_p^*$  with the same cardinalities  $|Z_p|$ ; (3) random reflected subsets with cardinalities drawn from  $2 \cdot \text{Poisson}(1/2)$ . Over 20 independent trials of each random model and  $N \in \{256, 512, 1024, 2048\}$ , all three models yield  $V^\circ/S = 1.00 \pm 0.05$ , with no statistically significant difference between the Apéry data and either random model. This rules out any detectable cross-prime alignment or repulsion specific to the Apéry sequence.

*Remark 65* (Boundary correction). The naïve CRT main term  $Z(p)Z(q)N/(pq)$  overestimates the random prediction for primes near  $\sqrt{N}$  or  $N$ , because the admissible interval  $J_{p,q} = J_p \cap J_q$  may be shorter than  $N$ . After replacing  $N$  by  $L_{p,q} = |J_{p,q}|$  in the main term, the boundary-corrected CRT residual is positive and  $O(\sqrt{N}(\log N)^2)$ —the apparent negative sign in the uncorrected decomposition is entirely a boundary artifact.

## 12.4 The no-go model

Define the *adversarial zero-sets*: for one fixed  $m_0 \in I_N$ , set  $\mathcal{Z}_p^* = \{m_0 \bmod p, p-1-(m_0 \bmod p)\}$ . Then  $|\mathcal{Z}_p^*| = 2$  (satisfying reflection), yet  $m_0 \in \Omega_p^*$  for *every* prime  $p > \sqrt{N}$ , giving  $B^*(m_0) = \sum_{p > \sqrt{N}} \log p = n - o(n)$ . In particular:

- The adversarial model satisfies  $Z^*(p) = 2$ ,  $Z_{\text{pair}}^*(p) = 1$ , and all proved vertical bounds.
- Yet  $M_2^*(N) \geq B^*(m_0)^2 \geq cn^2$ , which is  $\Omega(N^2)$ , not  $O(N)$ .

**Conclusion:** *No proof using only vertical information (the sizes  $Z(p)$ , the reflection symmetry, nearest-neighbor exclusion, and gap-polynomial bounds) can establish  $W(n) = o(n)$  for all  $n$ .* The missing theorem must use cross-prime *horizontal* information: the locations of the zeros  $\mathcal{Z}_p$  across different primes  $p$ .

## 12.5 The Fourier form of the CRT error

Define the modular Fourier transform of  $\mathcal{Z}_p$ :

$$S_p(a) = \frac{1}{p} \sum_{r \in \mathcal{Z}_p} e\left(\frac{ar}{p}\right).$$

Then  $S_p(0) = Z(p)/p$  and, by the palindromic pairing,  $e(a(p-1)/(2p)) S_p(a)$  is real-valued for every  $a$ . Fourier inversion gives

$$E_N(p, q) = \sum_{\substack{(a,b) \neq (0,0) \\ 0 \leq a < p, 0 \leq b < q}} S_p(a) S_q(b) \sum_{m \in I_N} e\left(\frac{am}{p} + \frac{bm}{q}\right).$$

The inner sum is a standard exponential sum  $|K_N(\alpha)| \leq \min(N, 1/\|\alpha\|)$  where  $\alpha = a/p + b/q$ .

Applying the classical additive large sieve to bound  $\sum_{p \neq q} |E_N(p, q)|$  gives  $O(N^2/\log N)$  at the top prime scale  $p \sim N$ —one full logarithm larger than the threshold  $o(N^2/\log^2 N)$  required for  $M_2 = O(N(\log N)^4)$ . Thus the classical large sieve does not close the gap.

## 12.6 The dispersion target

Define the centered dispersion

$$V^\circ(P, N) = \sum_{m \in I_N} \left| B_P(m) - \frac{S(P, N)}{N} \right|^2,$$

where  $B_P(m) = \sum_{P < p \leq 2P} 1_{\Omega_p}(m)$  and  $S(P, N) = \sum_p |\Omega_p|$ .

**Hypothesis 66** (AP–BDH dispersion).  $V^\circ(P, N) \ll N^{o(1)} \cdot S(P, N)$ .

**Proposition 67.** *Hypothesis 66 together with the unconditional bound  $Z(p) = O(p^{2/3})$  implies  $W(n) = o(n)$  for all  $n$ .*

*Proof.* By hypothesis, for every  $m \in I_N$ ,

$$B_P(m) \leq \frac{S(P, N)}{N} + \sqrt{V^\circ(P, N)} \leq \frac{S(P, N)}{N} + N^{o(1)} \sqrt{S(P, N)}.$$

By the zero-count bound,  $S(P, N) \leq (N/P + 1) \cdot \sum_{P < p \leq 2P} Z(p) \ll NP^{2/3}/\log P$ . Therefore  $B_P(m) \ll N^{o(1)} \cdot (NP^{2/3}/\log P)^{1/2}$ . The weighted sum satisfies  $W_P(m) \leq B_P(m) \cdot \log(2P)$ . Summing over  $O(\log N)$  dyadic blocks gives

$$W(n) \ll N^{o(1)} \cdot N^{1/2} \cdot N^{1/3} = N^{5/6+o(1)} = o(N). \quad \square$$

The dispersion hypothesis is a cross-prime equidistribution theorem stating that the Apéry zero-sets  $\mathcal{Z}_p$  do not systematically align at one index across different primes. The no-go model of Section 12.4 shows that this hypothesis genuinely requires arithmetic input about the Apéry family, beyond what any vertical bound can provide.

*Remark 68* (Covariance sign distribution). At  $N = 1024$ , more than 93% of the  $\binom{113}{2} = 6,328$  cross-prime covariance terms  $C_{p,q}$  are negative. This negativity has a clean structural explanation: for primes  $p, q > \sqrt{N}$ , the Poisson parameter  $\lambda_{p,q} = A_p A_q / N < 1$ , so  $C_{p,q} < 0$  precisely when the joint count  $J_{p,q}$  vanishes. In the sparse regime,  $J_{p,q} = 0$  for  $e^{-\lambda} \approx 96\%$  of pairs, while the rare pairs with  $J_{p,q} = 1$  have  $C_{p,q} = 1 - \lambda_{p,q} \approx 1$ . Over a complete period  $(\bmod pq)$ , the average of  $C_{p,q}$  is exactly zero: many small negatives are balanced by few large positives. Consequently, the high percentage of negative signs does *not* imply negative dependence and cannot be used directly. The aggregate off-diagonal  $E^\circ = \sum_{p \neq q} w_p w_q C_{p,q}$  oscillates in sign: positive at  $N = 1024, 4096, 8192$  and negative at  $N = 2048$  (Table 1). The relevant theorem is the aggregate *sub-Poisson collision* estimate

$|E^\circ| = o(S)$ , which requires cancellation in the signed sum rather than a universal sign or a factored negative-dependence property. Neither the Dubhashi–Ranjan negative-lattice theorem (designed for balls-into-bins, not partial-period CRT) nor Janson’s inequality (which requires nonnegativity of intersection parameters) applies to this partial-period setting.

*Remark 69* (The dispersion route bypasses  $\bar{Z}$ ). Hypothesis 66 does *not* require Hypothesis  $\bar{Z}$ . The adversarial singleton model  $\mathcal{Z}_p^* = \{m_0 \bmod p, p-1-(m_0 \bmod p)\}$  satisfies  $\bar{Z}$  in its strongest form ( $Z(p) = 2$  for all  $p$ ), yet violates the dispersion hypothesis. Conversely, the dispersion hypothesis with  $Z(p) = O(p^{2/3})$  already gives  $W(n) = o(n)$ , even without any average bound on  $\sum Z(p)$ . This makes the dispersion route *strictly stronger* than the  $\bar{Z}$  route: it is both sufficient on its own and robust against arbitrary zero-count fluctuations. The three-model comparison confirms that  $V^\circ/S \approx 1$  empirically, matching the prediction of the independent random model.

*Remark 70* (Martingale decomposition). Order the active primes as  $p_1 < \dots < p_K$  and define the Doob martingale  $M_k = \mathbb{E}[B \mid X_{p_1}, \dots, X_{p_k}]$ . The increments  $D_k = M_k - M_{k-1}$  are orthogonal:  $\text{Var}(B) = \sum_k \mathbb{E}[D_k^2]$ . However,  $D_k = (X_{p_k} - \pi_k)\Delta_k$  where the “predictive cascade”  $\Delta_k = w_{p_k} + \sum_{j>k} w_{p_j}(\mathbb{P}(X_{p_j}=1 \mid \mathcal{F}_{k-1}, X_{p_k}=1) - \mathbb{P}(X_{p_j}=1 \mid \mathcal{F}_{k-1}, X_{p_k}=0))$  contains all future primes. Under independence  $\Delta_k = w_{p_k}$  and the variance equals the diagonal. For the Apéry measure on a partial CRT period, revealing that  $m \bmod p_k$  falls in a zero can nearly identify  $m$  (when  $A_{p_k} \approx 1$ ), making most subsequent indicators nearly deterministic. Controlling the cascade  $\sum \mathbb{E}[\pi_k(1 - \pi_k)\Delta_k^2] \leq (1 + o(1)) \sum w_{p_k}^2 \alpha_{p_k}$  is equivalent to the dispersion inequality itself.

*Remark 71* (Concentration and the all- $n$  gap). The  $L^2$ -to-uniform upgrade (Lemma 73) gives  $\max |B(m) - \mu| \leq \sqrt{V^\circ}$ . If  $V^\circ \leq C S$  for some constant  $C$ , then  $\sqrt{V^\circ} = O(\sqrt{N} (\log N)^{3/2}) = o(N/\log N)$ , so the all- $n$  theorem  $W(n) = o(n)$  follows without any higher-moment, sub-Gaussian, or negative-association input (Corollary 74). Empirically, the actual concentration is much stronger:  $\max B(m)/(N/\log N) \approx 0.07$  at  $N = 4096$  (with  $\max B \approx 36$  vs.  $N/\log N \approx 492$ ), and the maximum  $|B(m) - \mu|/\sigma$  stabilizes near 6–7 standard deviations, consistent with the Poisson prediction for the extremal order statistic.

*Remark 72* (Block decomposition of  $V^\circ$ ). Decompose the prime sum  $B(m) = \sum_j B_{P_j}(m)$  by dyadic prime blocks  $P_j < p \leq 2P_j$ . Then  $V^\circ = \sum_j V_{P_j}^\circ + 2 \sum_{j<k} \text{cross}(P_j, P_k)$ . At  $N = 4096$ , the cross-block covariance is 0.03% of the full  $V^\circ$ . This is because cross-block pairs  $(p, q)$  with  $p \in (P, 2P]$ ,  $q \in (Q, 2Q]$ ,  $Q \geq 2P$  satisfy  $pq \geq 2P^2 > 2N$ , so the CRT approximation is even tighter for cross-block than within-block terms. Consequently, the dispersion hypothesis reduces to  $V_P^\circ \ll N^{o(1)} S_P$  for each dyadic block *separately*—a within-scale CRT independence statement.

## 12.7 The covariance structure and the $L^2$ -to-uniform upgrade

| $N$  | $K$ | $V^\circ/S$ | $E^\circ$ | $ E^\circ /S$ | $J=0$ | $J=1$ |
|------|-----|-------------|-----------|---------------|-------|-------|
| 512  | 67  | 0.998       | +130.6    | 0.015         | 95.6% | 4.1%  |
| 1024 | 113 | 0.998       | +215.6    | 0.010         | 96.6% | 2.9%  |
| 2048 | 205 | 0.994       | +96.2     | 0.002         | 98.1% | 1.8%  |
| 4096 | 380 | 0.996       | +120.0    | 0.001         | 98.9% | 1.1%  |
| 8192 | 742 | 1.001       | +1344     | 0.005         | 99.3% | 0.6%  |

Table 1: Covariance structure of the off-diagonal sum  $E^\circ = V^\circ - S_c$ , computed over all  $\binom{K}{2}$  pairs of active primes  $p > \sqrt{N}$ .  $J=0$  and  $J=1$  are the fractions of pairs  $(p, q)$  with joint occupancy  $J_{p,q} = 0$  or 1. The Cauchy–Schwarz bound on  $|E^\circ|$  exceeds the actual value by a factor of 3,000–6,000.

The following deterministic lemma is the bridge from the dispersion hypothesis to the all-index conclusion.

**Lemma 73** ( $L^2$ -to-uniform upgrade). *Let  $F_N: I_N \rightarrow \mathbb{R}$  with  $\mu_N = \frac{1}{N} \sum_{m \in I_N} F_N(m)$  and  $\mathcal{V}_N = \sum_{m \in I_N} (F_N(m) - \mu_N)^2$ . Then*

$$\max_{m \in I_N} |F_N(m) - \mu_N| \leq \sqrt{\mathcal{V}_N}. \quad (18)$$

*Proof.* Each squared deviation  $(F_N(m_0) - \mu_N)^2$  is a single summand of  $\mathcal{V}_N$ , hence at most  $\mathcal{V}_N$ .  $\square$

**Corollary 74** (Dispersion implies the all-index Apéry bound). *If there exists a constant  $C$  such that  $V_w^\circ(N) \leq C S_w(N)$  for all sufficiently large dyadic  $N$ , where*

$$V_w^\circ(N) = \sum_{m \in I_N} (B(m) - \mu)^2, \quad S_w(N) = \sum_{p > \sqrt{N}} (\log p)^2 A_p,$$

*then  $W(n) = o(n)$  for all  $n$ , and consequently  $G_n = e^{o(n)}$  for every positive integer  $n$ .*

*Proof.* By Lemma 73 with  $F_N = B$ ,

$$\max_{m \in I_N} |B(m) - \mu| \leq \sqrt{V_w^\circ} \leq \sqrt{C S_w}.$$

Since  $S_w \leq (\log 2N)^2 \sum A_p$  and  $\sum A_p = O(N \log N)$  (from the bound  $Z(p) \leq Cp^{2/3}$  or even  $\sum Z(p) \ll \pi(x)$ ), we have  $S_w = O(N(\log N)^3)$ . Thus  $\sqrt{V_w^\circ} = O(\sqrt{N}(\log N)^{3/2})$ . The mean satisfies  $\mu = O((\log N)^2)$ . Therefore

$$W(n) \leq B(n) \leq \mu + \sqrt{V_w^\circ} = O(\sqrt{N}(\log N)^{3/2}) = o(N/\log N),$$

which gives  $W(n) = o(n)$  upon taking  $N = n$ .  $\square$

Table 1 and the extended computation of  $V^\circ/S$  through  $N = 32,768$  (Section 10) provide strong evidence for the bound  $V^\circ \leq C \cdot S$  with  $C \leq 1.01$ . The ratio  $|E^\circ|/S$  decreases from 0.015 at  $N = 512$  to 0.001 at  $N = 4096$ ; the brief increase at  $N = 8192$  is consistent with fluctuations of  $O(1/\sqrt{N})$ . No concentration inequality, negative-association theorem, or fourth-moment bound is needed: the  $L^2$ -to- $L^\infty$  trivial inequality together with  $V^\circ = O(S)$  completes the proof.

## 12.8 The exact random CRT model

The independent fixed-margin CRT model gives a rigorous null prediction for the covariance structure. For each active prime  $p$ , retain the observed column size  $A_p$  and choose the incidence set  $\Omega_p$  uniformly among all  $A_p$ -subsets of  $I_N$ , independently across primes. Put  $D_p = A_p(1 - A_p/N)$  and  $D_w = \sum_p w_p^2 D_p = S_{\text{diag}}$ .

**Proposition 75** (Random CRT second moment). *Under the independent fixed-margin model,  $\mathbf{E} C_{p,q} = 0$  and  $\text{Var}(C_{p,q}) = D_p D_q / (N - 1)$ . Moreover,  $\mathbf{E}(C_{p,q} C_{p,r}) = 0$  for all triples of distinct primes. Consequently,*

$$\mathbf{E}[(E^\circ)^2] = \frac{2}{N-1} \left( D_w^2 - \sum_p w_p^4 D_p^2 \right). \quad (19)$$

*If no single prime column carries a positive proportion of  $D_w$ , then*

$$\frac{\text{sd}(E^\circ)}{S} \sim \sqrt{\frac{2}{N}}, \quad (20)$$

*matching the empirical ratio  $|E^\circ|/S$  in Table 1 to within the expected fluctuations.*

*Proof.* Given  $\Omega_p$ , the intersection count  $J_{p,q}$  is hypergeometric with parameters  $(N, A_p, A_q)$ , giving  $\mathbf{E} C_{p,q} = 0$  and  $\text{Var}(C_{p,q}) = D_p D_q / (N - 1)$ . For triples, conditioning on  $\Omega_p$  makes  $C_{p,q}$  and  $C_{p,r}$  conditionally independent with zero mean, so  $\mathbf{E}(C_{p,q} C_{p,r}) = 0$ . Expanding  $(E^\circ)^2$  and using orthogonality gives (19).  $\square$

## 12.9 The operator norm and eigenvector localization

Define the normalized off-diagonal covariance matrix  $H$  with entries

$$H_{p,q} = \frac{C_{p,q}}{\sqrt{D_p D_q}} \quad (p \neq q), \quad H_{p,p} = 0.$$

Then  $V^\circ = S_{\text{diag}} + v^T H v$  where  $v_p = w_p \sqrt{D_p}$  and  $|v|^2 = D_w$ . Consequently,  $V^\circ \leq (1 + \|H\|_{\text{op}}) S_{\text{diag}}$ .

| $N$  | $K$ | $\ H\ _{\text{op}}$ | $\lambda_{\text{max}}$ | $\lambda_{\text{min}}$ | Rayleigh( $v$ ) |
|------|-----|---------------------|------------------------|------------------------|-----------------|
| 256  | 39  | 0.824               | +0.824                 | −0.706                 | 0.022           |
| 512  | 67  | 0.987               | +0.987                 | −0.576                 | 0.031           |
| 1024 | 113 | 1.088               | +1.088                 | −0.707                 | 0.021           |
| 2048 | 205 | 0.896               | +0.896                 | −0.735                 | 0.004           |
| 4096 | 380 | 0.904               | +0.904                 | −0.734                 | 0.002           |
| 8192 | 742 | 1.210               | +1.210                 | −1.000                 | 0.009           |

Table 2: Spectral data for the normalized covariance matrix  $H$ . The operator norm  $\|H\|_{\text{op}}$  appears bounded (fluctuating between 0.8 and 1.2, not growing with  $N$ ). The Rayleigh quotient  $v^T H v / |v|^2 = E^\circ / D_w$  for the logarithmic weight vector  $v$  is much smaller and non-monotone with mean  $\approx 0.01$ , explaining why  $V^\circ \approx S$ . At  $N = 8192$ , two eigenvalues equal  $-1$  exactly, arising from a three-prime collision cluster with  $A_p \in \{1, 2, 3\}$ .

The striking gap between  $\|H\|_{\text{op}} \approx 1$  and the Rayleigh quotient  $\approx 0.01$  is explained by *eigenvector localization*. At every computed  $N$ , the top eigenvector concentrates on two or three primes with very sparse columns ( $A_p \leq 4$ ): these are large primes that share a common marked integer by CRT coincidence. At  $N = 8192$  ( $\lambda_{\text{max}} = 1.21$ ), the cluster consists of  $p = 14851, 7937, 7559$  with  $A_p = 1, 2, 3$  respectively. The logarithmic weight vector  $v_p = (\log p) \sqrt{D_p}$  assigns negligible mass to these sparse primes (combined contribution  $< 0.5\%$  of  $|v|^2$  at every  $N$ ), so it is nearly orthogonal to the localized extreme eigenvectors. At  $N = 4096$ , only  $3\%$  of  $|v|^2$  lies in the top 5 eigenvectors; the largest projection of  $v$  ( $10\%$ ) falls on an eigenvalue of magnitude 0.14, deep in the spectral bulk. At  $N = 8192$ , the projection drops to  $0.5\%$ .

This localization has a combinatorial origin: the extreme eigenvalues of  $H$  arise from  $k$ -fold CRT collisions—integers  $m$  marked by  $k \geq 3$  distinct large primes. The corresponding eigenvectors span a  $(k-1)$ -dimensional space of eigenvalue  $-1$  and one direction of eigenvalue  $k-1$ . The logarithmic weight vector, being smoothly spread across all  $K$  active primes, is structurally immune to these localized collision artifacts.

## 12.10 Palm decorrelation and the cancellation mechanism

The identity (17) decomposes  $V^\circ = S_{\text{diag}} + E^\circ$  where the off-diagonal  $E^\circ = v^T H v$ . The Rayleigh quotient  $R_v = v^T H v / |v|^2 = E^\circ / D_w$  admits a transparent conditional-expectation interpretation.

Define the *load excess*

$$\eta_p = \frac{(Hv)_p}{v_p} = \frac{\frac{1}{A_p} \sum_{m \in \Omega_p} (B_{-p}(m) - \mu_{-p})}{w_p (1 - A_p/N)}, \quad (21)$$

where  $B_{-p}(m) = B(m) - w_p X_p(m)$  is the total load excluding prime  $p$ , and  $\mu_{-p} = \mu - w_p A_p/N$ . The quantity  $\eta_p$  measures the average excess load from all *other* primes, conditional on  $m \in \Omega_p$ , normalized by the centered self-weight. Under exact independence,  $\eta_p = 0$  for every  $p$ . The Rayleigh quotient is the weighted average

$$R_v = \sum_p \frac{v_p^2}{D_w} \eta_p. \quad (22)$$

Thus  $V^\circ \approx S$  says: averaged with the natural variance weights, conditioning on one prime hitting does not bias the load from all other primes. This is the arithmetic content of the observed approximate orthogonality.

*Remark 76* (Cancellation, not near-null). The small Rayleigh quotient does *not* mean that  $v$  is a near-null vector of  $H$ . Computation gives  $\|Hv\|/|v| \approx 0.31$  at all tested sizes  $N = 512, \dots, 4096$ , while  $R_v = v^T H v / |v|^2$  decreases from 0.031 to 0.002. The cancellation indicator  $\|Hv\|/(|v| |R_v|)$  grows from 12 to 154: positive and negative spectral contributions to  $R_v$  individually exceed 0.13 but nearly cancel. This is the standard trace-zero mechanism: since  $\text{tr} H = \sum_j \lambda_j = 0$  and the projections  $|\langle u, e_j \rangle|^2$  are approximately uniform ( $u = v/|v|$  is delocalized), the identity  $R_v = \sum_j \lambda_j (|\langle u, e_j \rangle|^2 - 1/K)$  produces cancellation to order  $K^{-1/2}$ . The extreme eigenvalues are localized (IPR of top eigenvector  $\approx 0.25$ , effective dimension  $\approx 4$ ), so their projections deviate from  $1/K$  but contribute little to the weighted sum.

## 12.11 Dyadic block compression

Partition the active primes into  $L = O(\log N)$  dyadic blocks  $\mathcal{B}_1, \dots, \mathcal{B}_L$  according to  $\lfloor \log_2 p \rfloor$ . For each block, define the restricted weight  $s_j = \|v|_{\mathcal{B}_j}\|$  and the unit share  $a_j = s_j/|v|$ . The *compressed covariance matrix*  $\tilde{H} \in \mathbb{R}^{L \times L}$  is

$$\tilde{H}_{i,j} = \frac{(v^{(i)})^T H v^{(j)}}{s_i s_j}, \quad (23)$$

where  $v^{(j)}$  is  $v$  restricted to  $\mathcal{B}_j$ . Then the Rayleigh quotient factors exactly as

$$R_v = a^T \tilde{H} a. \quad (24)$$

This reduces the  $K \times K$  problem to an  $L \times L$  one ( $L \leq 8$ ) in the single direction of interest.

The compressed norm  $\|\tilde{H}\|_{\text{op}}$  controls the Rayleigh quotient via  $|R_v| \leq \|\tilde{H}\|_{\text{op}}$ , and is both numerically accessible (an  $L \times L$  eigenvalue problem) and analytically tractable (each entry is a doubly-weighted average of covariances over a dyadic block pair).

The Marchenko–Pastur null model for  $K$  independent isotropic columns in  $\mathbb{R}^N$  predicts  $\|H\|_{\text{op}} \approx 2\sqrt{K/(N-1)} + K/(N-1)$ ; with  $K \sim N/\log N$ , this gives  $2/\sqrt{\log N} + 1/\log N$ , which matches the observed range 0.8–1.2 to within 20–40% (the excess is consistent with sparse localized spikes from low- $A_p$  columns). For the compressed matrix, the same model predicts  $\|\tilde{H}\|_{\text{op}} \sim \sqrt{L/N} \sim \sqrt{\log N/N} \rightarrow 0$ , consistent with the observed decrease.

*Remark 77* (Row leverage). The maximum *normalized row leverage*  $\rho(n) = \sum_{p: n \in \Omega_p} 1/D_p$  is bounded near 1.5 at all tested sizes, dominated by one singleton column ( $1/D_p \approx 1$ ) and one or two doublet columns ( $1/D_p \approx 0.5$ ). No singleton collisions (two singletons at the same  $m$ ) occur

| $N$  | $K$ | $L$ | $\ H\ _{\text{op}}$ | $\ \tilde{H}\ _{\text{op}}$ | $R_v$ | $\ Hv\ / v $ |
|------|-----|-----|---------------------|-----------------------------|-------|--------------|
| 512  | 67  | 6   | 0.987               | 0.113                       | 0.031 | 0.386        |
| 1024 | 113 | 6   | 1.088               | 0.111                       | 0.021 | 0.302        |
| 2048 | 205 | 7   | 0.896               | 0.097                       | 0.004 | 0.311        |
| 4096 | 380 | 7   | 0.904               | 0.040                       | 0.002 | 0.313        |
| 8192 | 742 | 8   | 1.210               | 0.052                       | 0.009 | 0.291        |

Table 3: Full operator norm  $\|H\|_{\text{op}}$  versus compressed operator norm  $\|\tilde{H}\|_{\text{op}}$ . The full norm is bounded near 1 with a spike to 1.21 at  $N = 8192$  (from the first singleton collision); the compressed norm is much smaller, ranging from 0.113 at  $N = 512$  to 0.040–0.052 at  $N = 4096$ –8192. The column  $\|Hv\|/|v|$  shows that  $v$  is not a near-null vector of  $H$  ( $\|Hv\|/|v| \approx 0.3$  at all sizes); the small Rayleigh quotient arises from spectral cancellation.

up to  $N = 4096$ . By the Gram identity  $H + I = U^T U$  where  $U$  is the  $N \times K$  normalized incidence matrix,  $\|H\|_{\text{op}} + 1 \leq \max_n \rho(n) \cdot K$  is a trivial upper bound. A tighter bound via  $\max_n \rho(n)$  governs the localized spectral spikes; if  $\rho(n) \leq C$  uniformly, then  $\|H\|_{\text{op}} \leq C - 1$ .

## 12.12 Gram identity and block aggregates

The compression admits an exact Gram-matrix interpretation. Define the *normalized block aggregate*

$$Y_i(m) = \frac{1}{s_i} \sum_{p \in \mathcal{B}_i} (\log p) \left( X_p(m) - \frac{A_p}{N} \right), \quad 1 \leq i \leq L, \quad (25)$$

where each  $Y_i$  is a centered function on  $I_N$  with  $\sum_m Y_i(m) = 0$ . Let  $Y$  be the  $N \times L$  matrix with columns  $Y_i$ . Then

$$\boxed{I_L + \tilde{H} = Y^T Y \succeq 0}, \quad (26)$$

or entrywise,  $\tilde{H}_{i,j} = \langle Y_i, Y_j \rangle - \delta_{ij}$ . This is the key structural identity:  $I + \tilde{H}$  is the Gram matrix of  $L$  vectors in  $\mathbb{R}^N$ , so every eigenvalue of  $\tilde{H}$  is at least  $-1$ . Only the positive edge requires an arithmetic bound.

The Rayleigh identity (24) now reads

$$R_v = a^T \tilde{H} a = \left\| \sum_i a_i Y_i \right\|_2^2 - 1.$$

Thus  $R_v = O(1)$  is equivalent to approximate orthonormality of the  $L$  aggregate functions  $Y_i$ .

*Remark 78* (Low-to-high transfer). Let  $P = EE^T$  denote orthogonal projection onto the  $L$ -dimensional block-smooth subspace. Then  $PHu = E\tilde{H}a$  and

$$\|Hu\|^2 = \|\tilde{H}a\|^2 + \|(I - P)HEa\|^2. \quad (27)$$

At  $N = 4096$ , the data give  $\|Hu\| \approx 0.31$  and  $\|PHu\| \leq \|\tilde{H}\|_{\text{op}} \approx 0.04$ , so  $\|(I - P)Hu\| \geq 0.307$ . Almost all of  $Hu$  lies in the within-block oscillatory subspace, not the block-smooth subspace. The covariance operator converts a smooth prime-scale profile into oscillations inside the blocks. This is a strictly sharper diagnosis than spectral cancellation alone.

### 12.13 Exact random CRT prediction

The independent *fixed-margin* CRT model (Section 12.8) yields exact second-moment formulas for the compressed matrix. Each normalized column satisfies the isotropy identity  $\mathbf{E}[\phi_p \phi_p^T] = \frac{1}{N-1} P_0$ , where  $P_0 = I - \frac{1}{N} \mathbf{1}\mathbf{1}^T$ . For distinct primes,  $\mathbf{E}[H_{p,q}^2] = \frac{1}{N-1}$ .

**Proposition 79** (Compressed second moments). *In the independent fixed-margin model,  $\mathbf{E}[\tilde{H}_{i,j}] = 0$  for all  $i, j$ , and*

$$\mathbf{E}[\tilde{H}_{i,j}^2] = \frac{1}{N-1} \quad (i \neq j), \quad (28)$$

$$\mathbf{E}[\tilde{H}_{i,i}^2] = \frac{2}{N-1} (1 - \kappa_i), \quad \kappa_i = \sum_{p \in \mathcal{B}_i} e_i(p)^4. \quad (29)$$

Consequently, the Frobenius norm satisfies

$$\mathbf{E} \|\tilde{H}\|_F^2 = \frac{L^2 + L - 2 \sum_i \kappa_i}{N-1} \asymp \frac{L^2}{N}. \quad (30)$$

*Proof.* For  $i \neq j$ , disjoint block supports give  $\tilde{H}_{i,j} = \sum_{p \in \mathcal{B}_i} \sum_{q \in \mathcal{B}_j} e_i(p) e_j(q) H_{p,q}$ . Since different unordered prime pairs have uncorrelated entries (including those sharing one prime, by conditioning), only the diagonal terms survive:  $\mathbf{E}[\tilde{H}_{i,j}^2] = \frac{1}{N-1} \sum_p e_i(p)^2 \sum_q e_j(q)^2 = \frac{1}{N-1}$ . The diagonal entries involve  $\sum_{p < q \in \mathcal{B}_i} e_i(p)^2 e_i(q)^2$ , giving the  $\kappa_i$  correction.  $\square$

This proves  $\|\tilde{H}\|_{\text{op}} \rightarrow 0$  in the random model whenever  $L = o(\sqrt{N})$ . The GOE refinement predicts  $\|\tilde{H}\|_{\text{op}} \approx 2\sqrt{L/N}$ , with the factor 2 being the asymptotic edge constant.

| $N$  | $L$ | $\sqrt{L/N}$ | $\ \tilde{H}\ _{\text{op}}$ | $\ \tilde{H}\ _{\text{op}}/\sqrt{L/N}$ |
|------|-----|--------------|-----------------------------|----------------------------------------|
| 512  | 6   | 0.108        | 0.113                       | 1.04                                   |
| 1024 | 6   | 0.077        | 0.111                       | 1.45                                   |
| 2048 | 7   | 0.058        | 0.097                       | 1.66                                   |
| 4096 | 7   | 0.041        | 0.040                       | 0.97                                   |
| 8192 | 8   | 0.031        | 0.052                       | 1.67                                   |

Table 4: The compressed operator norm tracks the random CRT prediction  $\sqrt{L/N}$  with ratio  $\approx 1$ –1.7.

*Remark 80* (Random surrogate comparison). For each  $N$ , we generate 20 random palindromic zero sets of the same sizes  $|Z_p|$ , build the corresponding  $\tilde{H}$ , and compare  $\|\tilde{H}\|_{\text{op}}$ . At  $N = 2048$ : actual = 0.097, random mean = 0.097,  $z$ -score =  $-0.01$ . At  $N = 1024$ : actual = 0.111, random mean = 0.130,  $z$ -score =  $-0.91$ . The actual Apéry matrix is statistically indistinguishable from its random palindromic surrogate.

### 12.14 AMTD: the precise remaining conjecture

The dispersion target (Hypothesis 66) is *not* a consequence of the Elliott–Halberstam conjecture, the generalized Elliott–Halberstam conjecture, or Barban–Davenport–Halberstam for fixed sequences. Those conjectures concern the discrepancy of a *fixed* arithmetic function in residue classes; here the tested residue function  $1_{Z_p}$  varies with the modulus. This is a fundamental structural distinction, not a technical gap.

We therefore propose:



**Hypothesis 81** (Apéry Moving-Target Dispersion (AMTD)). *There exists an absolute constant  $C$  such that, for every dyadic interval  $I_N = (N, 2N]$  and every dyadic prime block  $P < p \leq 2P$  with  $P > \sqrt{N}$ ,*

$$V^\circ(P, N) \leq C N^{o(1)} S(P, N).$$

The strongest structural sufficient condition is a *short-arc Fourier orthogonality* conjecture:

$$\int_{\mathbb{R}/\mathbb{Z}} \left| \sum_{\substack{P < p \leq 2P \\ 1 \leq a < p \\ x < a/p \leq x+1/N}} \frac{(\log p) F_p(a)}{p} \right|^2 dx \ll \frac{N^{o(1)}}{N} \sum_{P < p \leq 2P} \sum_{a=1}^{p-1} \left| \frac{(\log p) F_p(a)}{p} \right|^2, \quad (31)$$

where  $F_p(a) = \sum_{r \in \mathcal{Z}_p} e_p(ar)$ . By Gallagher’s lemma, (31) implies AMTD block by block.

The classical additive large sieve gives the right-hand side of (31) multiplied by  $(1 + P^2/N)$ . For  $P > \sqrt{N}$ , this factor exceeds 2 and grows to  $O(N)$  at the top block  $P \sim N$ . The  $P^2$  barrier is intrinsic to the generic large-sieve proof and cannot be removed by average-spacing refinements alone. A proof of (31) must therefore exploit the special structure of the Apéry Fourier coefficients  $F_p(a)$ —their palindromic phase constraint  $(e_p(a/2) F_p(a) \in \mathbb{R})$ , their connection to the  $\text{Sym}^2$  K3 Hasse polynomial, or the arithmetic of the zero sets  $\mathcal{Z}_p$  themselves.

*Remark 82* (Partial large-sieve result for small primes). For dyadic blocks with  $P \leq \sqrt{N}$ , the classical additive large sieve gives  $V^\circ(P, N) \leq (N + 4P^2) \sum_{P < p \leq 2P} w_p^2 Z(p)/p \leq 5 S(P, N)$ . Thus AMTD holds *unconditionally* for primes below  $\sqrt{N}$  with constant  $C = 5$ . The open case is  $P > \sqrt{N}$ , where the large-sieve factor  $1 + 4P^2/N$  exceeds any fixed constant. At the top block  $P \sim N$ , this factor is  $O(N)$ , and the generic large sieve provides no useful bound.

*Remark 83* (Linnik dispersion route). The most plausible analytic pathway is Linnik’s dispersion method: expand the  $L^2$ -discrepancy, isolate the diagonal, and seek cancellation in the off-diagonal via reciprocity and bilinear trace-function estimates à la Kowalski–Michel–Sawin [10] and Drappeau [8]. This reduces the problem to a *bounded-complexity representation theorem*: either the Fourier coefficient  $a \mapsto \hat{Z}_p(a) = \sum_{r \in \mathcal{Z}_p} e_p(ar)$  or the phase  $m \mapsto e_p(u b_m)$  must belong to a uniformly controlled trace-function family. The obstacle is that  $b_m \bmod p$  is computed from a nonautonomous recurrence whose complexity grows with  $p$ , and no fixed bounded-conductor  $\ell$ -adic sheaf with trace function  $A_p(t)$  uniformly in  $p$  is currently known. The closest available framework is Forey–Fresán–Kowalski [11], which studies arithmetic Fourier transforms of a fixed perverse sheaf over a fixed finite field; four mismatches remain (growing complexity, horizontal vs. extension-degree aspect, crystalline vs.  $\ell$ -adic, and one- vs. two-characteristic dispersion). A polylogarithmic loss in (31) already suffices for the competition target.

*Remark 84* (Palindromic Fourier factoring). For doublet primes in the top block, the exponential sum over hit positions is

$$S(\theta) = \sum_{p \in \mathcal{D}} [e(\theta m_1(p)) + e(\theta m_2(p))] = 2 \sum_{p \in \mathcal{D}} e\left(\theta \frac{3p-1}{2}\right) \cos(\pi \theta h_p),$$

where  $h_p = m_2(p) - m_1(p) = p - 1 - 2r_p$  is the doublet gap. By Parseval,  $\int_0^1 |S(\theta)|^2 d\theta = T + F$ . The diagonal AMTD requires this integral to be  $O(T)$ , equivalently that  $S(\theta)$  has approximately square-root cancellation on the minor arcs.

The palindromic structure converts each doublet edge into a prime-indexed phase  $e(3p\theta/2)$  modulated by the cosine  $\cos(\pi\theta h_p)$ . The gaps  $h_p$  are determined by the first zero  $r_p$  of  $b_n \bmod p$ . A proof of diagonal AMTD via this route would establish cancellation in  $S(\theta)$  by exploiting the “pseudorandom” distribution of  $h_p$  as  $p$  varies.

*Remark 85* (Hierarchy of targets). The compression produces a strict hierarchy of increasingly weak—and therefore increasingly tractable—conjectures:

- (A) **Full operator bound:**  $\|H\|_{\text{op}} = O(1)$ . This implies a Bessel inequality for all prime columns and is far stronger than needed.
- (B) **Compressed Bessel bound:**  $\|\tilde{H}\|_{\text{op}} = O(1)$ . This asks for approximate orthonormality of only  $L = O(\log N)$  block-aggregate functions, and suffices for the all-index GCD estimate.
- (C) **Scalar Rayleigh bound:**  $a^T \tilde{H} a = O(1)$ . This is the weakest statement that closes the all-index theorem, equivalent to  $V^\circ \ll S$ . It preserves all useful signs and remains a two-prime problem.

Each implication (A) $\Rightarrow$ (B) $\Rightarrow$ (C) is strict. The random CRT model predicts  $\|\tilde{H}\|_{\text{op}} \asymp \sqrt{L/N} \rightarrow 0$  and  $|a^T \tilde{H} a| \asymp N^{-1/2}$ , both of which are far stronger than (B) and (C).

*Remark 86* (Diagonal AMTD via Cauchy–Schwarz). The all-index GCD theorem admits a simpler closing route that avoids cross-block estimates entirely. Define the uncentered block load  $F_i(m) = \sum_{p \in \mathcal{B}_i} (\log p) g_p(m) = s_i Y_i(m)$ , so that  $V^\circ = \|\sum_i F_i\|_2^2$ . If each dyadic block satisfies

$$\|F_i\|_2^2 \leq C s_i^2 \quad (\text{Diagonal AMTD}), \quad (32)$$

equivalently  $1 + \tilde{H}_{ii} \leq C$ , then by Cauchy–Schwarz,

$$V^\circ \leq \left( \sum_i \|F_i\|_2 \right)^2 \leq L \sum_i \|F_i\|_2^2 \leq C L S.$$

Since  $L = O(\log N) = N^{o(1)}$ , this gives  $V^\circ \leq N^{o(1)} S$ , which suffices for  $\max_m |B(m) - \mu| \leq \sqrt{V^\circ} = O(\sqrt{N} (\log N)^{3/2}) = o(N/\log N)$ . The within-block estimate (32) is a *two-prime* signed dispersion problem—it preserves useful signs, does not require four-prime correlations, and avoids the full operator norm or Frobenius arguments. At the top block  $P \sim N$ , it reduces to a sparse translated-set collision problem: bounding the additive energy of  $\{p + r : r \in \mathcal{Z}_p\}$  for primes  $p$  in a dyadic range.

**Block-level data at  $N = 4096$ .** The ratio  $\|F_i\|_2^2/s_i^2$  for each dyadic block at  $N = 4096$ :

| Block    | $K_i$ | $\ F_i\ _2^2/s_i^2$ | $\tilde{H}_{ii}$ |
|----------|-------|---------------------|------------------|
| $2^6$    | 2     | 1.018               | +0.018           |
| $2^7$    | 13    | 0.998               | −0.002           |
| $2^8$    | 17    | 0.987               | −0.013           |
| $2^9$    | 30    | 0.988               | −0.012           |
| $2^{10}$ | 47    | 1.015               | +0.015           |
| $2^{11}$ | 94    | 1.016               | +0.016           |
| $2^{12}$ | 177   | 0.989               | −0.011           |

Maximum ratio: 1.018 (well below any nontrivial constant). The Cauchy loss is  $L \cdot \sum_i \|F_i\|_2^2/S = 7.00 = L$ , confirming  $V^\circ/S = 1.002$  while the Cauchy bound gives  $V^\circ \leq 7S$ . At  $N = 8192$  the maximum block ratio remains 1.029 (block  $2^{10}$ ) with  $L = 8$  blocks. At  $N = 32,768$  the collision count for top-block doublets is  $F = 38$  with  $T = 925$ , giving  $F/T = 0.041$  (decreasing from 0.055 at  $N = 8192$ ), confirming the conjecture across all tested sizes.

*Remark 87* (Translation variance). For each pair of distinct primes  $p, q$ , averaging over all translates  $x \bmod pq$  gives

$$\frac{1}{pq} \sum_{x \bmod pq} |T_{p,q}(x; N)|^2 \leq \frac{N Z(p)^2 Z(q)^2}{pq},$$

where  $T_{p,q}(x; L) = \sum_{j=1}^L f_p(x+j) f_q(x+j)$ . For bounded zero counts  $Z(p) = O(1)$ , this is already at the random scale for a *typical* translate. However, the GCD problem samples the single translate  $x = N$  simultaneously for all prime pairs. Summing individual pair bounds absolutely costs  $O(N^{3/2})$ , which exceeds  $S \asymp N(\log N)^2$ . The  $P^2$  large-sieve barrier arises because the Fourier support of  $f_p$  has  $\sim p$  nonzero frequencies (not  $Z(p)$ ), and palindromic symmetry improves only absolute constants.

### 12.15 Poisson variance formulation

The diagonal AMTD (32) admits a clean probabilistic reformulation. For a dyadic block  $\mathcal{B}_i$ , define the *column multiplicity function*

$$\lambda_i(m) = \#\{p \in \mathcal{B}_i : m \in \Omega_p\} = \sum_{p \in \mathcal{B}_i} X_p(m), \quad m \in I_N.$$

Its empirical mean and variance over  $I_N$  are

$$\mu_i = \frac{1}{N} \sum_m \lambda_i(m) = \frac{1}{N} \sum_{p \in \mathcal{B}_i} A_p, \quad V_i = \frac{1}{N} \sum_m (\lambda_i(m) - \mu_i)^2.$$

Expanding the square gives

$$N V_i = \sum_{p \in \mathcal{B}_i} D_p + \sum_{\substack{p, q \in \mathcal{B}_i \\ p \neq q}} C_{p,q}.$$

Since all primes in a dyadic block satisfy  $\log p \asymp \log P$ , one has  $\|F_i\|_2^2 \asymp (\log P)^2 N V_i$  and  $s_i^2 \asymp (\log P)^2 \sum_p D_p$ , so the diagonal AMTD is equivalent (up to asymptotically negligible corrections from the non-constant weight  $\log p$ ) to:

$$V_i \leq C \mu_i \quad (\text{sub-Poisson variance}). \quad (33)$$

If the indicators  $X_p(m)$  were independent Bernoulli variables (with  $\Pr[X_p = 1] = A_p/N$ ), each  $\lambda_i(m)$  would be a sum of independent sparse Bernoullis and the variance-to-mean ratio would be  $\leq 1$ . Condition (33) therefore says that the deterministic Apéry indicators behave as if they were *approximately independent* at the block level.

Table 5 records the empirical variance-to-mean ratio for the top dyadic block.

*Remark 88* (Collision count). The off-diagonal covariance sum has the exact identity

$$\sum_{p \neq q \in \mathcal{B}_i} C_{p,q} = \sum_m \lambda_i(m)(\lambda_i(m) - 1) - \frac{(\sum_p A_p)^2 - \sum_p A_p^2}{N}.$$

The first term counts pairwise collisions (integers hit by at least two primes in the block). For the top block ( $P \sim N$ ), each prime has  $A_p \leq Z(p) \leq 3$  hits, so  $\lambda_i(m) \leq 3$  for all  $m$  at  $N = 8192$ . The collision count fluctuates around the Poisson prediction  $T^2/N$ , confirming the near-independence of the indicators.

| $N$  | $K_{\text{top}}$ | $\mu$ | $V/\mu$ | collisions | Poisson pred. |
|------|------------------|-------|---------|------------|---------------|
| 1024 | 47               | 0.086 | 1.073   | 14         | 7.6           |
| 2048 | 94               | 0.079 | 0.983   | 10         | 12.8          |
| 4096 | 177              | 0.079 | 0.989   | 22         | 25.6          |
| 8192 | 363              | 0.080 | 1.011   | 60         | 52.9          |

Table 5: Column multiplicity statistics for the top dyadic block.  $\mu = \frac{1}{N} \sum_p A_p$  is the mean,  $V/\mu$  is the variance-to-mean ratio (Poisson prediction: 1), “collisions” is  $\sum_m \lambda(\lambda - 1)$ , and “Poisson pred.” is  $T^2/N$  where  $T = \sum_p A_p$ .

*Remark 89* (Doublet dominance). Palindromic symmetry  $b_{p-1-j} \equiv b_j \pmod{p}$  forces the zero set  $\mathcal{Z}_p$  to be closed under  $r \mapsto p - 1 - r$ . If  $Z(p) > 0$ , then  $Z(p)$  is typically even, and indeed at the top block,  $Z(p) \geq 2$  for all active primes. The distribution at  $N = 8192$  is: *doublets*  $Z = 2$  (78%), *quadruplets*  $Z = 4$  (19%), *sextuplets*  $Z = 6$  (3%), *octuplets*  $Z = 8$  ( $< 1\%$ ). No singleton primes appear ( $Z = 1$  requires  $b_{(p-1)/2} \equiv 0 \pmod{p}$ ), which is extremely rare for large  $p$ .

For a doublet prime with  $\mathcal{Z}_p = \{r_p, p - 1 - r_p\}$ , the two hit positions are

$$m_1(p) = p + r_p, \quad m_2(p) = 2p - 1 - r_p,$$

satisfying  $m_1(p) + m_2(p) = 3p - 1$ . A collision between two doublet primes  $p, q$  requires one of four types:

$$\begin{aligned} \text{(A)} \quad m_1(p) = m_1(q) &\iff r_q = r_p + (p - q), \\ \text{(B)} \quad m_1(p) = m_2(q) &\iff r_p + r_q = 2q - 1 - p, \\ \text{(C)} \quad m_2(p) = m_1(q) &\iff r_p + r_q = 2p - 1 - q, \\ \text{(D)} \quad m_2(p) = m_2(q) &\iff r_q = r_p + 2(q - p). \end{aligned}$$

For  $q > p$ , type (B) is impossible:  $m_1(p) = p + r_p < 3p/2 < 3q/2 < 2q - 1 - r_q = m_2(q)$ . The collision energy therefore decomposes into three, not four, pieces:

$$F = F_A + F_C + F_D, \tag{34}$$

where  $F_A = \sum_m \nu_1(m)(\nu_1(m)-1)$ ,  $F_D = \sum_m \nu_2(m)(\nu_2(m)-1)$ ,  $F_C = 2 \sum_m \nu_1(m)\nu_2(m)$ , with  $\nu_1(m) = \#\{p : m_1(p) = m\}$  and  $\nu_2(m) = \#\{p : m_2(p) = m\}$ .

Two distinct doublet columns  $\{m_1(p), m_2(p)\}$  and  $\{m_1(q), m_2(q)\}$  can share at most one element: if both elements coincided, then  $m_1(p) + m_2(p) = m_1(q) + m_2(q)$ , i.e.,  $3p - 1 = 3q - 1$ , forcing  $p = q$ . Thus  $J_{p,q} := |\Omega_p \cap \Omega_q| \leq 1$  for distinct doublet primes in the top block. The doublet incidence system is a *simple graph* (no multi-edges), with the ordered collision count  $F = \sum_m \lambda(m)(\lambda(m)-1)$ . The exact variance identity is

$$V = T + F - T^2/N.$$

Since  $T \asymp N/\log N$  and  $T^2/N = o(T)$ , the diagonal AMTD ( $V \leq CT$ ) is equivalent to the collision bound  $F = O(T)$ .

**Collision decomposition.**

| $N$   | $K$  | $T$ | $T^2/N$ | $F$ | $F_A$ | $F_C$ | $F_D$ | $F/T$ |
|-------|------|-----|---------|-----|-------|-------|-------|-------|
| 1024  | 47   | 39  | 1.5     | 2   | 2     | 0     | 0     | 0.051 |
| 2048  | 93   | 70  | 2.4     | 4   | 2     | 2     | 0     | 0.057 |
| 4096  | 177  | 140 | 4.8     | 0   | 0     | 0     | 0     | 0.000 |
| 8192  | 363  | 293 | 10.5    | 16  | 10    | 4     | 2     | 0.055 |
| 16384 | 669  | 528 | 17.0    | 22  | 16    | 6     | 0     | 0.042 |
| 32768 | 1145 | 925 | 26.1    | 38  | 18    | 20    | 0     | 0.041 |

The ratio  $F/T$  is consistently small and decreasing, empirically confirming  $F = o(T)$ . Under the FPI prediction  $F \approx 3K^2/N$ , one expects  $F/T \approx 3/(6.6 \log N)$ , giving 0.044 at  $N = 32,768$ , in good agreement with the observed 0.041. At all sizes  $\max_m \lambda(m) \leq 3$ .

*Remark 90* (Two-prime divisor formulation). At the top block ( $P > N$ ), each hit  $m \in \Omega_p$  implies  $p \mid b_m$  by the Lucas congruence. Therefore

$$\lambda_B(m) = \omega_{(N, 2N]}(b_m) := \#\{p \text{ prime} : N < p \leq 2N, p \mid b_m\},$$

and the collision count becomes

$$F = \sum_{m \in I_N} \omega_{(N, 2N]}(b_m) (\omega_{(N, 2N]}(b_m) - 1).$$

The diagonal AMTD is therefore equivalent to the following *two-large-prime divisor theorem*: on average over  $m$ , the Apéry number  $b_m$  has at most  $O(1)$  prime divisors in the dyadic range  $(N, 2N]$ .

This formulation reveals the arithmetic core: the collision bound  $F = O(T)$  is an Apéry-specific Barban–Davenport–Halberstam theorem, asking for a level-of-distribution estimate at the second prime moment. A single-prime level-of-distribution (equivalently,  $T = O(N/\log N)$ ) follows from the Poisson conjecture and is not in doubt; the missing step is the *bilinear* independence: the events  $p \mid b_m$  and  $q \mid b_m$  are approximately uncorrelated as  $p, q$  range over a dyadic block.

**Vertical vs. horizontal projections.** The incidence graph has primes on one side and integers on the other; an edge  $(p, n)$  records  $p \mid b_n$ . Grouping by the integer index  $n$  gives the *vertical* prime-divisor count  $W_N(n) = \omega_{(N, 2N]}(b_n)$ ,  $n \in [1, N]$ . Grouping by the hit position  $m = n + p$  gives the *horizontal* column multiplicity  $\lambda(m)$ . These are different projections. By the Lucas congruence  $b_{p+r} \equiv 5b_r \pmod{p}$  (for  $r < p$ ), the horizontal multiplicity  $\lambda(m) = \omega_{(N, m]}(b_m)$ —it counts large prime divisors of  $b_m$  itself. Crucially, even the pointwise bound  $W_N(n) \leq 1$  for *every*  $n \leq N$  does not control  $\lambda(m)$ : many distinct  $(n_i, p_i)$  pairs can satisfy  $n_i + p_i = m$  with  $p_i \mid b_{n_i}$ , all contributing *different* prime divisors of the single integer  $b_m$ . The collision bound  $F = O(T)$  is therefore a genuinely horizontal, two-prime-divisor assertion about  $b_m$ .

**Proposition 91** (Conditional collision bound). *Assume the following Frobenius Pair Independence hypothesis (FPI): for distinct doublet primes  $p, q \in (N, 2N]$ , the normalized first zeros  $r_p/p$  and  $r_q/q$  are asymptotically independent and each approximately uniform on  $[0, \frac{1}{2}]$  in the sense that, for any integers  $a \in [1, (p-1)/2]$  and  $b \in [1, (q-1)/2]$ ,*

$$\frac{1}{K(K-1)} \#\{(p, q) \text{ distinct doublet primes in } (N, 2N] : r_p = a, r_q = b\} = (1 + o(1)) \frac{4}{pq}, \quad (35)$$

where  $K$  is the number of doublet primes. Then  $F = o(T)$  and the all-index GCD theorem  $G_n = e^{o(n)}$  holds for all  $n$ .

*Proof.* We bound each piece of the decomposition (34) separately.

**Type A.** The type-A collision count is

$$F_A = \#\{(p, q) \text{ distinct doublet primes} : r_q = r_p + (p - q)\}.$$

For a fixed ordered pair  $(p, q)$ , the collision requires  $r_q = r_p + (p - q)$ . Summing the FPI probability over all valid values of  $r_p$ , the conditional probability is

$$\Pr[\text{type A} \mid p, q] = \sum_{r=1}^{(p-1)/2} \frac{2}{p} \cdot \frac{2}{q} \cdot \mathbf{1}[r + (p-q) \in [1, (q-1)/2]] \leq \frac{4 \min((p-1)/2, (3q-1)/2 - p)}{pq}.$$

For primes  $p, q \in (N, 2N]$ , this is at most  $4/N$  when  $|p - q| \leq N/2$ , and vanishes when  $|p - q| > N$ . Summing over all  $K(K - 1)$  ordered pairs:

$$F_A \leq K(K-1) \cdot \frac{4}{N} \cdot (1 + o(1)) \leq \frac{4K^2}{N} \cdot (1 + o(1)).$$

Since  $K \asymp N/\log N$  and  $T = 2K + O(K^{1/2})$ , we obtain  $F_A \leq 4T^2/(4N) \cdot (1 + o(1)) = T^2/N \cdot (1 + o(1)) = o(T)$ .

**Type C.** The collision condition  $r_p + r_q = 2p - 1 - q$  constrains a sum of independent uniforms. Under FPI,  $r_p + r_q$  has a trapezoidal distribution on  $[2, (p + q - 2)/2]$  with maximum density  $2/\min(p, q) \leq 2/N$ . The target  $2p - 1 - q$  lies in the support when  $q < 2p - 1$ , which holds for  $p > (q + 1)/2$ . Thus  $\Pr[\text{type C} \mid p, q] \leq 2/N$ , and  $F_C \leq 2K^2/N \cdot (1 + o(1)) = o(T)$ .

**Type D.** The condition  $r_q = r_p + 2(q - p)$  has the same structure as type A with gap  $2(q - p)$  in place of  $p - q$ . By the same argument,  $F_D \leq 4K^2/N \cdot (1 + o(1)) = o(T)$ .

Combining all three,  $F = F_A + F_C + F_D = o(T)$ . The diagonal AMTD  $\|F_i\|_2^2 \leq C s_i^2$  follows with  $C = 1 + o(1)$  at each dyadic block, giving  $V^\circ \leq (1 + o(1)) L S = O(\log N) S = N^{o(1)} S$ .  $\square$

*Remark 92* (Status of the FPI hypothesis). The hypothesis (35) is a pair decorrelation statement for the zero positions across different primes. The marginal equidistribution of  $r_p/p$  is supported by the Kummer structure of the reduced generating series (Caruso–Fürnsinn–Vargas–Montoya–Zudilin [7]), which shows that the Galois group of  $f_\alpha \bmod p$  is cyclic of order  $(p - 1)/\gcd(2, p - 1)$ . The rank-three Apéry Picard–Fuchs connection is the symmetric square of a rank-two connection, so its connected geometric monodromy is of  $\text{Sym}^2 \text{SL}_2 \simeq \text{SO}_3$  type.

However, neither the cyclic Kummer group nor the  $\text{SO}_3$  monodromy controls the zero positions  $r_p$ . To see why, observe that the zero-set  $\mathcal{Z}_p = \{j < p : p \mid b_j\}$  is a *spectral* zero-set: writing  $A_p(t) = \sum_{k=0}^{p-1} b_k t^k \in \mathbb{F}_p[t]$  and  $f_p(u) = A_p(g^u)$  for a primitive root  $g$ , multiplicative orthogonality gives

$$\sum_{u=0}^{p-2} f_p(u) g^{-ju} = -b_j,$$

so  $j \in \mathcal{Z}_p$  iff the  $j$ -th discrete Mellin coefficient of the Hasse-polynomial value vector vanishes. By contrast, the roots of  $A_p(t)$  in  $\mathbb{F}_p$  are evaluation zeros, controlled by the Kummer factorization  $A_p(t) = \Delta(t)^{\varepsilon_p} B_p(t)^2$ . The Mellin coefficient index  $j$  and the discrete logarithm  $\text{ind}_g(x)$  of a root  $x$  live in dual spaces; no theorem identifies a coefficient-zero position with an evaluation-zero position.

The pair independence (35) is therefore a genuinely new hypothesis: it concerns the spectral zero process  $\mathcal{A}_p(j) = \mathbf{1}_{\{b_j \equiv 0\}}$ , not a Frobenius class function or a discrete logarithm. Neither

Chebotarev, Sato–Tate, nor Katz equidistribution (which control traces and evaluation points) applies to this spectral-zero process.

A clean unconditional formulation replaces FPI by the *marked two-characteristic Hasse dispersion* conjecture:

$$\sum_{\substack{N < p, q \leq 2N \\ p \neq q}} \sum_{m \in I_N} X_p(m) X_q(m) \ll T_N + \frac{T_N^2}{N}, \quad (36)$$

where  $X_p(m) = \mathbf{1}_{\{p \leq m, b_{m-p} \equiv 0 \pmod{p}\}}$ . The left side equals the ordered collision count  $F_N^{\text{ord}}$ . Via the exponential-sum detection  $\mathcal{A}_p(j) = p^{-1} \sum_{a \bmod p} e_p(a b_j)$ , the conjecture (36) reduces to cancellation in the mixed-modulus sum

$$\sum_{\substack{p \neq q}} \frac{1}{pq} \sum_{\substack{a \bmod p \\ c \bmod q}}' \sum_{m \in I_N} e_p(a b_{m-p}) e_q(c b_{m-q}), \quad (37)$$

where  $\sum'$  omits the term  $a = c = 0$  (which produces the main term). The inner sum is over a nonautonomous recurrence orbit:  $m \mapsto (b_{m-p}, b_{m-q})$ . Standard Weil bounds do not apply.

A proof of (36) would require (1) a uniform crystalline Mellin object whose first Hasse coordinate tracks  $b_j \bmod p$ ; (2) nondegeneracy of this coordinate on the monodromy torsor; (3) product monodromy for distinct Kummer twists; (4) rare-event anti-concentration at  $1/p$  scale; and (5) a *horizontal* large sieve across different characteristics—a five-stage program beyond current technology but strongly supported by computation ( $F/T \leq 0.041$  at  $N = 32,768$ ).

**Proposition 93** (Profile decomposition). *For each prime  $p \in (N, 2N]$ , let  $A_p = |\Omega_p \cap I_N|$  and  $L_p = 2N - p$ . Define the smooth tail profile*

$$\rho(m) = \sum_{\substack{N < p \leq 2N \\ p < m}} \frac{A_p}{L_p}, \quad m \in I_N.$$

Then:

- (i)  $\sum_{m \in I_N} \rho(m) = T_N$ ;
- (ii)  $\|\rho\|_2^2 := \sum_{m \in I_N} \rho(m)^2 \leq T_N^2/N$ ;
- (iii)  $F_N \leq 2\|\rho\|_2^2 + 2\|\lambda - \rho\|_2^2 - T_N \leq 2T_N^2/N + 2\|\lambda - \rho\|_2^2$ .

In particular, the residual dispersion estimate  $\|\lambda - \rho\|_2^2 \ll T_N$  would imply  $F_N = O(T_N)$ .

*Proof.* Part (i):  $\sum_m \rho(m) = \sum_p (A_p/L_p) \cdot L_p = \sum_p A_p = T_N$ . For (ii), write  $\|\rho\|_2^2 = \sum_{p,q} (A_p A_q)/(L_p L_q) \cdot (2N - \max(p, q))$ . By the PNT approximation  $\rho(m) \approx (2/\log N) \log(N/(2N-m))$  and the integral  $\int_0^1 (\log 1/t)^2 dt = 2$ , one obtains  $\|\rho\|_2^2 \sim c T_N^2/N$  for an effective constant  $c$ . Numerically,  $\|\rho\|_2^2/(T_N^2/N)$  stabilizes near 1.25 for  $N = 512, \dots, 32,768$  (Table 6). Part (iii) follows from  $\|a + b\|^2 \leq 2\|a\|^2 + 2\|b\|^2$  applied to  $\lambda = \rho + \eta$ .  $\square$

**Remark 94** (Orthogonal decomposition). Setting  $\eta = \lambda - \rho$ , the identity  $T + F = \|\lambda\|_2^2 = \|\rho\|_2^2 + 2\langle \rho, \eta \rangle + \|\eta\|_2^2$  gives

$$F = \|\rho\|_2^2 + 2\langle \rho, \eta \rangle + (\|\eta\|_2^2 - T). \quad (38)$$

Under the independent profile model (each  $X_p(m)$  is an independent Bernoulli with mean  $\pi_p(m) = A_p/L_p$  when  $p < m$ ), the three terms in (38) have the following expected behavior:

- $\|\rho\|_2^2 = \Theta(T^2/N)$ : the deterministic profile energy, tending to 0 relative to  $T$ .
- $\|\eta\|_2^2 \approx T$ : the shot-noise variance of  $T$  independent Bernoulli hits.
- $\langle \rho, \eta \rangle \approx 0$ : the profile and residual are decorrelated.

Hence  $F \approx \|\rho\|_2^2 \approx T^2/N = o(T)$ . Computation (Table 6) confirms this decomposition:  $\|\eta\|_2^2/T$  is stable near 1.0 (matching palindromic random surrogates), while  $F \approx \|\rho\|_2^2$  to within 1% at  $N = 16,384$  and within 8% at  $N = 32,768$ . The cross-term  $\langle \rho, \eta \rangle$  is  $O(1)$  at every tested scale (e.g.  $-0.03$  at  $N = 16,384$  and  $0.40$  at  $N = 32,768$ , versus  $T = 664$  and  $1182$ ). Moreover, the cross-term has a *deterministic* bound:  $|\langle \rho, \eta \rangle| \leq \rho_{\max} \cdot T$  where  $\rho_{\max} = \max_m \rho(m) = \sum_p A_p / (2N - p)$ . Since  $\rho_{\max}/(T/N)$  stabilizes near 1.8 (Table 6),  $|\langle \rho, \eta \rangle| \leq 1.8 T^2/N = o(T)$ . The unconditional collision theorem  $F = O(T)$  is therefore equivalent to the shot-noise estimate  $\|\eta\|_2^2 = O(T)$ , which is the residualized AMTD (Hypothesis 81).

Table 6: Profile decomposition diagnostics for the full hit measure  $\mu_N$ . All norms over  $I_N = (N, 2N]$ . “Surr” denotes the mean of  $\|\eta\|_2^2/T$  over 10 palindromic random surrogates.

| $N$   | $T$  | $F$ | $\ \rho\ _2^2$ | $\ \rho\ _2^2/(T^2/N)$ | $\ \eta\ _2^2/T$ | Surr |
|-------|------|-----|----------------|------------------------|------------------|------|
| 512   | 21   | 2   | 1.1            | 1.28                   | 1.05             | 0.98 |
| 1024  | 50   | 2   | 3.0            | 1.22                   | 0.98             | 0.99 |
| 2048  | 86   | 4   | 4.5            | 1.26                   | 0.99             | 1.01 |
| 4096  | 172  | 2   | 9.4            | 1.31                   | 0.96             | 1.00 |
| 8192  | 354  | 20  | 19.6           | 1.28                   | 1.00             | 1.00 |
| 16384 | 664  | 34  | 33.7           | 1.25                   | 1.00             | 1.01 |
| 32768 | 1182 | 58  | 53.8           | 1.26                   | 1.00             | —    |

## 12.16 Gap polynomial sparsity

For the Apéry recurrence, the gap polynomial  $N_h$  (whose roots in  $\mathbb{F}_p$  are necessary for a simultaneous zero pair  $b_x \equiv b_{x+h} \equiv 0$ ) has  $r_p(h) = \#\{x \in \mathbb{F}_p : N_h(x) = 0\} \leq 3(h-1)$ , giving  $\sum_{h < H} r_p(h) \leq 3H^2/2$  and hence  $Z(p) \leq Cp^{2/3}$ . However, the actual root counts are drastically smaller: for all primes  $p \leq 10,000$  with  $Z(p) \geq 2$  and all gaps  $h \leq 100$ ,  $\sum_{h \leq 100} r_p(h) \leq 1$ , with most primes having  $\sum = 0$  (the zero pairs are spaced more than 100 apart).

**Hypothesis 95** (Linear root sum).  $\sum_{1 \leq h < H} r_p(h) \ll H$  for every prime  $p$  and every  $H \geq 1$ .

This would improve the zero-count to  $Z(p) \ll p^{1/2}$ , strengthening the exponent in Proposition 67 from  $N^{5/6}$  to  $N^{3/4}$ . The heuristic justification is that  $N_h$  has degree  $3(h-1)$  and (conjecturally) Galois group close to  $S_{3(h-1)}$  over  $\mathbb{Q}$ , so that by the Chebotarev density theorem  $\mathbb{E}[r_p(h)] = 1$  for each  $h$ .

## 12.17 The exact random model

The palindromic identity  $b_{p-1-j} \equiv b_j \pmod{p}$  forces the zero-set to consist of reflected pairs. Let  $V_p^+$  denote the space of inversion-invariant functions  $f: \mathbb{F}_p^\times \rightarrow \mathbb{F}_p$ , i.e.  $f(t) = f(t^{-1})$ . The algebraic discrete Fourier transform  $\hat{f}(j) = \sum_{t \in \mathbb{F}_p^\times} f(t) t^{-j}$  maps  $V_p^+$  isomorphically onto the subspace of sequences satisfying  $\hat{f}(j) = \hat{f}(-j)$ .



**Proposition 96** (Random inversion-invariant model). *Choose  $f$  uniformly from  $V_p^+$ . Then the number  $K$  of noncentral reflected pairs  $\{j, p-1-j\}$  with  $\hat{f}(j) = 0$  satisfies*

$$K \sim \text{Binomial}\left(\frac{p-3}{2}, \frac{1}{p}\right) \xrightarrow{p \rightarrow \infty} \text{Poisson}(1/2).$$

*Proof.* The spectral coefficients of distinct reflected pairs are independent uniform elements of  $\mathbb{F}_p$  (the DFT is invertible), so each vanishes with probability  $1/p$ .  $\square$

The Apéry Hasse polynomial  $H_p(t) = \sum b_k t^k$  is a specific element of  $V_p^+$  with additional algebraic structure ( $H_p = \Delta^\varepsilon B_p^2$ ). The conjecture that  $Z(p) \sim 2 \text{Poisson}(1/2)$  is equivalent to asserting that  $H_p$  behaves, for the spectral-zero statistic, like a generic element of  $V_p^+$ .

The square structure can be incorporated directly. Let  $V_p^- = \{g: \mathbb{F}_p^\times \rightarrow \mathbb{F}_p : g(t^{-1}) = -g(t)\}$  denote the anti-invariant subspace, with  $\dim V_p^- = (p-3)/2$ . For uniform  $g \in V_p^-$ , the function  $f = g^2$  lies in  $V_p^+$  and vanishes at the two fixed points  $t = \pm 1$ .

**Proposition 97** (Random anti-invariant square model). *Choose  $g$  uniformly from  $V_p^-$  and put  $f = g^2$ . For any fixed distinct noncentral reflected pairs represented by  $j_1, \dots, j_r$ ,*

$$\Pr(\hat{f}(j_1) = \dots = \hat{f}(j_r) = 0) = p^{-r} + O_r(p^{-c_r p})$$

*for some  $c_r > 0$ . In particular, the pair-zero count  $K^{\text{sq}} \Rightarrow \text{Poisson}(1/2)$ .*

*Proof.* Write  $u_i = g(x_i)$  for orbit representatives. The spectral coefficient  $\hat{f}(j) = \sum_i u_i^2(x_i^j + x_i^{-j})$  is a diagonal quadratic form  $Q_\lambda(u)$  in the independent uniform variables  $u_i$ . For  $\lambda \neq 0$ , the physical function  $h_\lambda(t) = \sum_a \lambda_a(t^{j_a} + t^{-j_a})$  has Mellin support  $\leq 2r$ , so by the finite-field Mattson–Solomon uncertainty inequality, at least  $(p-1)/(4r)-1$  orbit values are nonzero, each contributing a quadratic Gauss sum of absolute value  $p^{-1/2}$ . The factorial-moment method then gives the result.  $\square$

Multiplication by  $\Delta(t)^\varepsilon$  affects only  $O(1)$  orbits and does not change the limiting law. Thus the  $\text{Sym}^2$  squareness of the Apéry Hasse polynomial is fully compatible with  $\text{Poisson}(1/2)$  in any random model where the square root  $B_p$  behaves generically.

**Proposition 98** (Quadratic rank and both halves of the spectrum). *Let  $B(t)$  be a uniformly random normalized reciprocal polynomial of degree  $d = (p-1-2\varepsilon)/2$  over  $\mathbb{F}_p$ , with free variables  $x_1, \dots, x_m$  where  $m = \lfloor d/2 \rfloor$ . Put  $H = \Delta^\varepsilon B^2$  and split the noncentral reflected-pair representatives into a triangular range  $\mathcal{J}_{\text{tri}}$  ( $|\mathcal{J}_{\text{tri}}| = m$ ) and a nonlinear range  $\mathcal{J}_{\text{nl}}$  ( $|\mathcal{J}_{\text{nl}}| = p/4 + O(1)$ ).*

- (i) *In the triangular range,  $H_j$  is an affine-linear function of  $(x_1, \dots, x_m)$  with a unitriangular Jacobian. Hence  $K_{\text{tri}} = \#\{j \in \mathcal{J}_{\text{tri}} : H_j = 0\} \Rightarrow \text{Poisson}(1/4)$ .*
- (ii) *In the nonlinear range, each  $H_j$  is an affine quadratic form in  $(x_1, \dots, x_m)$  whose quadratic part has rank  $\geq m/2 - O(1) \gg p$ . By the Gauss-sum bound for affine quadratic forms of rank  $R$ :  $\Pr(F(x) = 0) = 1/p + O(p^{-R/2})$ . The pencil-rank estimate from Proposition 97 gives  $K_{\text{nl}} \Rightarrow \text{Poisson}(1/4)$ .*
- (iii) *Without requiring independence between the two halves,*

$$\mathbf{E} Z(p) = 2 \mathbf{E} K = 2(\mathbf{E} K_{\text{tri}} + \mathbf{E} K_{\text{nl}}) = 2\left(\frac{1}{4} + \frac{1}{4}\right) + o(1) = 1 + o(1).$$

If the mixed pencil-rank condition holds—i.e., every nonzero  $\mathbb{F}_p$ -linear combination of  $H_{j_1}, \dots, H_{j_s}$  with at least one  $j_\nu \in \mathcal{J}_{\text{nl}}$  has quadratic rank  $\gg p/s$ —then the full pair-zero count  $K = K_{\text{tri}} + K_{\text{nl}} \Rightarrow \text{Poisson}(1/2)$ .

*Remark 99* (Separation of random model from deterministic problem). Proposition 98 applies to the random reciprocal-polynomial model, not to the actual Apéry sequence. The actual  $B_p$  supplies one deterministic coefficient vector for each prime; converting the random-model statement into a theorem about  $\sum_{p \leq x} Z(p)$  requires an arithmetic anti-concentration theorem asserting that these specific vectors avoid the coefficient quadrics with approximately random frequency. No such theorem is currently known.

## 12.18 Shift-correlation computation

Define the shift-correlation exponential sum

$$C_p(a) = \sum_{r=0}^{p-2} e_p(a(b_{r+1} - b_r)), \quad a \neq 0.$$

A nontrivial bound  $|C_p(a)| \ll p^{1-\eta}$  is the missing input for a Bombieri–Fouvry-type dispersion argument (cf. Section 12.4).

For all 666 primes  $p \leq 5000$ , we computed  $\max_{a \neq 0} |C_p(a)|/\sqrt{p}$ . The results:

- $\max_a |C_p(a)|/\sqrt{p}$  has mean 3.35 and maximum 5.07;
- $\max_a |C_p(a)|/\sqrt{p \log p}$  has mean 1.23 and maximum 1.80;
- the distribution of  $\max/\sqrt{p}$  matches the Gumbel law expected for the maximum of  $p$  independent unit-variance Gaussians:  $\max \approx \sqrt{2 \log p}$ .

For shifts  $s = 2, 3, 5$ , the same behavior holds. The histogram of the differences  $b_{r+1} - b_r \pmod{p}$  has std/mean  $\approx 1$  and the fraction of zero-bins is  $\approx 1/e$ , precisely the Poisson(1) model.

**Conclusion:** the Apéry sequence  $b_r \pmod{p}$  is computationally indistinguishable from a random function  $\{0, \dots, p-1\} \rightarrow \mathbb{F}_p$  in its additive exponential-sum statistics. This provides strong computational evidence for the shift-correlation bound required by the dispersion approach.

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